

Unit 3 - The Fabric of the Internet and AI

▼ Teacher resources

▼ Printing Options

This unit explores how the Internet and networks function, their societal and ethical impacts, and their real-world applications. You examine data transmission, IP addressing, DNS resolution, network security, and AI's role in cybersecurity. Through hands-on simulations and discussions, you analyze how data moves through networks, how protocols ensure secure communication, and how issues like the digital divide affect global connectivity. Ethical problem-solving and collaboration help you propose solutions to improve Internet access and security. Grounded in computational thinking and real-world relevance, this unit fosters curiosity, critical thinking, and innovation, preparing you to engage with networking technologies and their societal implications.

► By the end of this unit, students will be able to ...

Finished Teaching This Unit?

Answer this [short survey](#) to let the Code.org curriculum team know how the unit went.

Week 1	Lesson 1: Introduction to the Internet	Lesson 2: Sending Bits	Lesson 3: IP Addresses	Lesson 4: Routers and Redundancy	Lesson 5: Packets
Week 2	Lesson 6: HTTP and DNS	Lesson 7: The Internet's Hidden Barriers	Lesson 8: Internet Access Advocacy Campaign Project		
Week 3	Lesson 9: Network Security	Lesson 10: Emerging Technologies	Lesson 11: Network Security Risk Assessment Project		

Key	☐ Instructional Lesson	Assessment	Unplugged Lesson
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View calendar

Show All Lessons

Hide All Lessons

▼ Pre-Assessment

▼ Pre-Assessment

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 1-3

Pre-Assessment



▼ Content, Part 1

▼ Lesson 1: Introduction to the Internet

How does the Internet connect billions of devices across the globe?

Students are introduced to the ways in which physical components like routers, switches and servers work with physical and wireless connections in order to transmit data globally. Students use an AI chatbot to simulate what it is like for devices to send and receive data reliably over the Internet. Finally, students are introduced to a new widget: The Internet Simulator, which they will use throughout this unit to explore the inner workings of the Internet.

 1-3

AI Networks Chat



 4

Internet Simulator

▼ Lesson 2: Sending Bits

How are messages broken into bits and sent across networks?

This lesson engages students in a hands-on simulation of network communication, where they act as network devices to explore how data is transmitted efficiently and accurately. Through two structured challenges, students construct and optimize networks using strings, transmit binary messages using hand signals and cards, and experience challenges like delays, congestion, and

errors in transmission. The lesson concludes with a discussion on how real-world networks address these challenges using protocols like TCP to ensure reliable data transfer.

This lesson contains no levels.

▼ Lesson 3: IP Addresses

Why do devices on the Internet need addresses?

In this lesson, students explore how IP addresses enable communication across networks by acting as unique identifiers for devices. They trace the flow of data through a network, identify issues caused by duplicate IP addresses, and examine how AI optimizes network traffic and reliability.



1-5

IP Addressing and Visualizing

1

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▼ Lesson 4: Routers and Redundancy

How does data find its way across the Internet and how does redundancy improve reliability?

This lesson engages students in hands-on exploration of how routers direct data and how redundancy ensures network reliability. Through using an Internet Simulator and diagramming network paths, students analyze router logs and participate in discussions that connect real-world detours to data transmission, deepening their understanding of network resilience.



1-2

Internet Simulator and Networks Diagram Analysis

1

2

▼ Lesson 5: Packets

How does data travel reliably to reach its destination on an unreliable network?

In this lesson, students explore how data is divided into packets for transmission and the challenges of ensuring reliable delivery. Through an unplugged simulation and an interactive Internet Simulator activity, students experience real-world issues like packet loss and delays. They develop and refine protocols to improve reliability. The lesson concludes with a discussion on TCP and UDP in order to highlight the tradeoffs networks face between reliability and speed.



Internet Simulator: Packets



Revising with AI



Check for Understanding

▼ Lesson 6: HTTP and DNS

How do websites find their way to your browser using DNS and HTTP?

This lesson explores how DNS (Domain Name System) and HTTP (HyperText Transfer Protocol) work together to connect devices across the Internet. Students engage in hands-on activities, including an IP address collection exercise, a DNS query simulation, and an HTTP request exploration, to understand how devices locate websites and retrieve web content. Through discussions, chatbot interactions, and network monitoring exercises, students analyze how DNS translates domain names into IP addresses and how HTTP requests enable seamless access to online resources.



Internet Simulator: DNS Queries



DNS Hierarchy Chatbot



AI Chat: Energy Costs of the Internet



HTTP Requests Exploration

4

5

6

▼ Lesson 7: The Internet's Hidden Barriers

Who has access to the Internet and who is left behind?

This lesson explores the hidden barriers to Internet access, focusing on the digital divide and net neutrality. Students analyze real-world data from Pew Research to identify disparities in Internet access and engage in a case study simulation where they interact with personas facing digital accessibility challenges. Through discussions, data interpretation, and chatbot interactions, students connect these issues to education, employment, and civic participation, considering solutions to promote equitable Internet access.



1-5

Pew Research Center: Internet Access

1

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6

Choose a Persona

▼ Mid-Unit Project

▼ Lesson 8: Internet Access Advocacy Campaign Project

Career Journeys

This project-based lesson guides students in developing an advocacy campaign addressing digital equity, the digital divide, and net neutrality. Students explore real-world challenges, analyze case studies, and propose actionable solutions using AI-driven technology and policy recommendations. Through research, collaboration, and peer feedback, students refine their message and create a compelling advocacy campaign.



1

AI Message Testing



2-3

Final Project Reflection

▼ Content, Part 2

▼ Lesson 9: Network Security

Career Journeys

What vulnerabilities exist in networks, and what strategies can mitigate their impact?

In this lesson, students take on the role of cybersecurity analysts to investigate real-world network security threats and identify effective defenses. Through an interactive simulation, they explore vulnerabilities such as phishing, weak passwords, default credentials, and social engineering. After investigating each vulnerability, students propose a mitigation strategy that can be applied in a real-world context.

 1

Identify, Defend, Reflect!

 2

Check for Understanding

 3

Student Survey

▼ Lesson 10: Emerging Technologies

Career Journeys

What makes AI effective at enhancing network security, and how does it compare to traditional methods?

This lesson explores how AI enhances cybersecurity by detecting and responding to threats in real time. Through a hands-on simulation and discussions, students analyze how AI compares to traditional security methods and consider the challenges of implementing AI-driven solutions. The lesson encourages students to think critically about AI's benefits, limitations, and role in network security.



1

Threat Detection Chatbot



2

Check for Understanding



3

Student Survey

▼ End of Unit Project

▼ Lesson 11: Network Security Risk Assessment Project

Career Journeys

This project-based lesson guides students through conducting a Network Security Risk Assessment, identifying vulnerabilities, and proposing AI-driven security solutions. Students analyze real-world cybersecurity risks, categorize them, and assess their societal and ethical implications before refining their recommendations through peer review. By the end of the project, students communicate their findings in a professional risk assessment report.



1

Company Case Study Choices



2

AI-Driven Solutions



3

AI Chat Evaluation

4-5

4

5



6

End of Unit Survey

▼ End of Unit Assessment and Survey

▼ End of Unit Assessment

This lesson is locked - you need to become a verified teacher to unlock it. [Learn more.](#)

 1-3

End of Unit Assessment



▼ End of Unit Survey

 1-3

End of Unit Survey



Lesson 1: Introduction to the Internet

45 minutes

Overview

How does the Internet connect billions of devices across the globe?

Students are introduced to the ways in which physical components like routers, switches and servers work with physical and wireless connections in order to transmit data globally. Students use an AI chatbot to simulate what it is like for devices to send and receive data reliably over the Internet. Finally, students are introduced to a new widget: The Internet Simulator, which they will use throughout this unit to explore the inner workings of the Internet.

Standards

Full Course Alignment

CSTA K-12 Computer Science Standards (2017)

- ▶ **AP** - Algorithms & Programming
- ▶ **CS** - Computing Systems
- ▶ **NI** - Networks & the Internet

CSTA K-12 Computer Science Standards (2026)

- ▶ **SYS** - Systems and Security

Agenda

Warm Up (10 minutes)

Activity (30 minutes)

AI Networks Chat

Explore the Internet Simulator

Wrap Up (5 minutes)

Objectives

Students will be able to:

- Describe the basic infrastructure of the Internet
- Explain how devices communicate over the Internet

Preparation

- Test out the Internet Simulator. If you open up the level on two different tabs, you can use it by yourself.
- Print copies of the AI Networks Chat activity guide (one for each student).

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the students

- **AI Networks Chat** - Activity Guide
▼ Make a Copy
- **Fabric of the Internet and AI Unit Guide** - Activity Guide
▼ Make a Copy

Vocabulary

- **Internet tower** - a piece of infrastructure that sends and receives wireless signals
- **router** - a type of computer that forwards data across a network
- **server** - a powerful computer that stores and delivers websites, files,

and messages when requested by a user

- **switch** - a component that helps send data to the right place inside a local network

Teaching Guide

Warm Up (10 minutes)

 **Discuss:** *What components might be needed to connect two devices across the world?*

 **Discussion Goal:** Students should draw on their prior knowledge and ideas about technology to brainstorm potential tools or systems needed for global communication. The goal is to get them thinking about the overall infrastructure (e.g., "a connection," "a way to direct messages") that enables communication between devices. Students might identify components such as a WiFi or a cell tower, or even a satellite as ones needed to connect two devices.

 **Distribute:** Give each student their own copy of the The Fabric of the Internet and AI Unit Guide. Students will use this resource throughout the unit.

☒ **Do This:** Define *router* and *switch*.

☒ **Do This:** Define *internet towers* and *servers*.

 **Discuss:** Click through the animated slide to display the prompts.

- *How might these components (routers, switches, servers) work together to connect devices across the world?*
- *What might happen if one of these components fails during communication?*

 **Discussion Goal:** Students identify experiences with using networking components like routers to connect to the internet, and by extension, to other devices. They also likely make connections to at least one of those components failing and subsequently losing their connection. Making these connections, and introducing the names of the components, will create a shared vocabulary and understanding as the foundation for the lesson.

☒ **Do This:** Introduce the lesson objectives.

☒ **Do This:** Introduce the Question of the Day.

Activity (30 minutes)

AI Networks Chat (20 minutes)

 **Distribute:** Give each student a copy of the *AI Networks Chat* activity guide.

☒ **Do This:** Demonstrate how to get basic information using the AI chatbot.

☒ **Do This:** Demonstrate using more specific prompts in order to get more helpful information.

☒ **Do This:** Demonstrate prompts that make connections between topics in order to show relationships.

☒ **Do This:** Click through the animated slide to display troubleshooting strategies for AI chatbot interactions.

☒ **Do This:** Group students into pairs.

 **Transition:** Direct students to Lesson 1, Level 1.

- **Level 1:** Students interact with an AI Chatbot to complete their activity guide.
- **Level 2:** Students check the accuracy of AI chat.
- **Level 3:** Students complete a check for understanding.



1-3

AI Networks Chat

1

2

3

 **Circulate:** As students are interacting with the chatbot and completing their activity guides, ask them questions about the responses they are getting and checking for understanding.



Assessment Opportunity ▲

Students' answers in Level 3 can be used as a formative assessment of their understanding of the function of each network component.

Activity guides can also be collected and used to formatively assess student understanding of network components.

Explore the Internet Simulator (10 minutes)

 **Transition:** Prepare students to explore The Internet Simulator.

☒ **Do This:** Show students how to make a connection with their partner.



Teaching Tip ▲

Keep it quick! Give a **brief** demo using the next two slides, then let students explore on their own. Bring them back for a discussion after they have explored for a few minutes.



Remarks

When you connect, you'll see a visual representation of the direct connection between you and your partner. You'll also see some options for changing how data is represented in the My Device tab.

☒ **Do This:** Show students the "My Device" tab and demonstrate how you can turn on or off the different representations of data (binary, decimal, ASCII).

☒ **Do This:** Introduce the different sections of the Internet Simulator widget for sending messages and reviewing message logs.

☒ **Do This:** Direct students to join their partners on Level 4 in order to explore the Internet Simulator tool.



4

Internet Simulator

Circulate: Check that students are sending and receiving messages, and ask them to describe what they notice in the logs. Get students to start thinking about how the Internet Simulator compares to the real thing by asking questions such as, “What happens when you try to send a message?” or “How does the simulator compare to the real Internet?”

Teaching Tip

Keep this exploration around five minutes long. Students have more chances to explore and interact in later lessons—this is just a first look!

Discuss: *How is the Internet Simulator similar to the Internet? How is it different?*

Discussion Goal: Encourage students to speak from their own experience of using the Internet. Students might suggest that it is similar to the Internet because they can send information to another person and it all comes back to zeroes and ones. They might suggest it is different from the Internet because it can only send text and it takes a couple seconds to send a text instead of instantly.

Wrap Up (5 minutes)

Do This: Review the learning objectives.

Do This: Place students in pairs or small groups of three.

Do This: Direct students to briefly share their thoughts about how the Internet connects devices across the globe before individually writing their own response to the Question of the Day in their Unit Guide.

Teaching Tip

Encourage students to go beyond basic definitions by making connections to today's activities.

Assessment Opportunity

The Question of the Day effectively assesses the lesson objectives if students provide responses that explain:

- The basic infrastructure of the Internet, including key components like routers, switches, servers, and physical / wireless connections
- An example from the lesson such as how a request for a website travels from a user's device to a server and back

Discuss: *How does the Internet connect billions of devices across the globe?*

Discussion Goal: Students should recognize that the Internet connects devices through a global network of routers, switches, and servers that direct and transmit data across wired and wireless connections. They should also recognize that data takes different forms (ASCII, decimal, binary).



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Lesson 2: Sending Bits

45 minutes

Overview

How are messages broken into bits and sent across networks?

This lesson engages students in a hands-on simulation of network communication, where they act as network devices to explore how data is transmitted efficiently and accurately. Through two structured challenges, students construct and optimize networks using strings, transmit binary messages using hand signals and cards, and experience challenges like delays, congestion, and errors in transmission. The lesson concludes with a discussion on how real-world networks address these challenges using protocols like TCP to ensure reliable data transfer.

Standards

Full Course Alignment

CSTA K-12 Computer Science Standards (2017)

- **NI** - Networks & the Internet

CSTA K-12 Computer Science Standards (2026)

- **SYS** - Systems and Security

Agenda

Warm Up (5 minutes)

Activity (35 minutes)

Wrap Up (5 minutes)

Objectives

Students will be able to:

- Describe how binary data is transmitted across a network
- Explain how network structure and protocols improve the reliability of data transmission

Preparation

- Print and cut out binary message cards (ensure they are in the correct order)
- Prepare strings for each group (refer to the group size chart for correct amounts)
- Ensure students have scratch paper and pencils for recording bits
- Review the Teaching Guide Handout to understand student roles and key takeaways

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the students

- **Role Cards for Challenge 1 and 2** - Resource [▼ Make a Copy](#)
- **The Internet: Wires, Cables, and WiFi** - Video

Vocabulary

- **Transmission Control Protocol (TCP)** - a protocol that ensures that all parts of a message are received and ordered correctly

Teaching Guide

Warm Up (5 minutes)

🎬 **Video:** Show the video "The Internet: Wires, Cables, & Wifi". Encourage students to think about how computers convert data into signals as they watch the video.

💡 Teaching Tip

The video will stop at 2:53. As students watch the video, encourage them to think about how computers convert data into signals. After watching the video, students should recognize that computers convert digital data into signals for transmission. Encourage students to think about the ways in which devices are able to communicate accurately, and the complexity of global communication.

💬 **Discuss:** *Why is binary such an effective way for computers to send and receive information?*

- Use Think-Pair-Share to facilitate this discussion.

💡 **Discussion Goal:** Help students to recognize that binary aligns with physical states (simple on/off, light/dark) that are easy to detect reliably. Students may mention that binary transmission reduces complexity, minimizes errors, and can be universally adopted for data transmission.

💡 Teaching Tip

If students struggle to respond, ask guiding questions, such as "What happens if a signal is weak or noisy? Why is binary better in these cases?" or "Can you think of other systems that use only two states (like light switches, Morse code)?"

☑ **Do This:** Introduce the lesson objectives.

☑ **Do This:** Introduce the Question of the Day.

Activity (35 minutes)

☑ **Do This:** Put students into groups of 5-7.

📄 **Distribute:** Give each group of students a set of data cards (to be set aside and used with Network Challenge 2) and string according to the following chart:

- Groups of 5: 10 strings
- Groups of 6: 15 strings

- Groups of 7: 21 strings

🔗 Teaching Tip ▲

String Calculation: The number of strings per group is determined by adding up all numbers less than the group size.

Example:

- Group of 5: Needs $4+3+2+1 = 10$ strings
- Group of 8: Needs $7+6+5+4+3+2+1 = 28$ strings

☑ **Do This:** Introduce the activity by explaining that students will act as network devices to simulate how bits (1s and 0s) are transmitted over a network. Emphasize that they must follow the instructions carefully to ensure accurate transmission.

🔗 Teaching Tip ▲

Refer to the Teaching Guide - Human Network Bit Transmission Challenge handout to gain a complete understanding of how the Network and Transmission Challenges should work, how to set them up, and key takeaways.

☑ **Do This:** Explain that each string represents a direct connection between two devices. Each student can have multiple connections but can only be directly linked to another person through one string at a time.

☑ **Do This:** Have students create a fully connected network by linking every person to every other person with strings.

🔄 **Circulate:** Ensure each student has direct connections to everyone else in the group. Look for students noticing inefficiencies and encourage them to share observations.

🔗 Teaching Tip ▲

Allow students to talk and work together to discover what their fully connected network should look like. Provide guidance to any groups who are struggling. See the Teaching Guide - Human Network Bit Transmission Challenge handout for an example of what this should look like.

Students will need to maintain their connections to engage in the transmission challenge on next slide.

👁 Assessment Opportunity ▲

Observe if students recognize that having direct connections to everyone is inefficient. Listen for discussions about how difficult it would be to scale this model to a large network.

☑ **Do This:** Once groups have their fully connected network, instruct them to assign roles. Explain what each role involves and provide the Sender with the Transmission Challenge 1 card so they know what bits they need to send.

📄 **Distribute:** Give Challenge 1 Role Cards to each student in the group and review the assigned task for each role.

☑ **Do This:** Once groups have their assigned roles, ensure that the sender and receiver are ready to transmit and receive messages.

💡 Teaching Tip ▲

Check that students understand their roles and responsibilities before starting the activity on the next slide.

☑ **Do This:** Have the sender transmit a full message (HEY) using binary while the receiver writes it down.

🔄 **Circulate:** Encourage Network Monitors to be actively engaged—they should note if a bit is dropped, arrives late, or is recorded incorrectly. Observe whether students fall behind or misinterpret bits. If errors occur, discuss why protocols are needed to correct mistakes.

💡 Teaching Tip ▲

Encourage students to use the metronome on the slide to keep transmission time consistent, transmitting one bit by holding up a 1 or 0 every second. If students miss a bit or get confused, don't correct them right away—this will help highlight the need for protocols like TCP later.

☑ **Do This:** Have groups translate their binary message to ASCII using the table.

🔄 **Circulate:** Look for groups who get the wrong translation and ask them where the error occurred in transmission.

💡 Teaching Tip ▲

If students notice errors in their message, have them reflect: How could an automated system detect and correct errors? This sets up the need for protocols like TCP.

💬 **Discuss:** Click through the animated slide to display the prompts.

- *What did you notice about this fully connected network?*
- *Would it be realistic for the Internet to work this way?*

💡 **Discussion Goal:** Students should recognize:

- **Advantage:** Transmission was **quick** and **direct** since every device was connected.
- **Disadvantages:**
 - **Not scalable**—too many direct connections make large networks impractical.
 - **Still prone to errors**—even with direct paths, mistakes in transmission can still occur.
 - **Real networks don't work like this**—instead of every device connecting directly, **the Internet relies on routers and paths to relay messages efficiently.**

☑ **Do This:** Explain that in this second challenge, the same rules apply: each string represents a direct connection between two devices. Each student can have multiple connections but can only be directly linked to another person through one string at a time.

☑ **Do This:** Explain that in this second challenge, students must remove unnecessary connections while still ensuring that everyone can communicate indirectly. Have students reduce their connections, ensuring that even if one string is cut, all students can still communicate.

🔄 **Circulate:** Look for groups struggling to maintain connectivity. Ask: If you remove too many connections, will everyone still receive data?

Teaching Tip

Emphasize that this reflects how real networks work—we don't connect every device directly; instead, we rely on routers and pathways. Encourage students to strategically place their connections.

Do This: Once groups have built their more efficient network, instruct them to assign roles. Explain what each role involves and provide the Sender with the bit cards for "hi" so they know what to send—one card per second.

Distribute: Give Challenge 2 Role Cards to each student in the group and review the assigned task for each role.

Do This: Review the Sender, Router, and Receiver roles using the slide. Ensure the Sender has the bit cards in the correct order.

Teaching Tip

Remind students that routers pass only to others not holding a card and should only pass one card at a time—but let some students break this rule to simulate congestion. Use that confusion to build a bridge to real-world delays and packet handling

Do This: Instruct students to send the binary message through the network using the "not busy" rule for routers. Ensure each bit card is passed at a rate of one per second using the metronome for pacing.

Circulate: Observe if routers get backed up (this mimics real-world Internet congestion) so that you can discuss this during the activity discussion debrief.

Teaching Tip

Encourage students to use the metronome on the slide to keep transmission time consistent, transmitting one card every second. Remind students to only hold one card at a time, however, it is okay to allow students (routers) to hold too many bits as this creates network congestion and is fine for students to experience this firsthand.

Do This: Have groups check if they successfully transmitted the message by translating the binary to ASCII using the table.

Circulate: Look for groups who get the wrong translation and encourage them to reflect on where the error occurred in transmission.

Teaching Tip

If errors occur, have students identify where the transmission failed—did a router delay a bit? Did packets arrive out of order? Relate this to real-world network errors—like when a video buffers or an image loads slowly on a website.

Discuss: Click through the animated slide to display the discussion prompts.

- *How does the network setup compare to the real Internet?*
- *What delays occurred? How does this reflect real-world networks?*

💡 **Discussion Goal:** Students should understand:

- Real networks face delays, congestion, and packet loss.
- Messages may arrive out of order, requiring reassembly (similar to how TCP works).
- More efficient networks still experience bottlenecks.

🗣️ **Discuss:** *What is one way we might check if some bits were missing or not received in the right order?*

- Use Think-Pair-Share to facilitate this discussion.

💡 **Discussion Goal:** Students should identify error-checking methods (like sequence numbers), resending missing data if a bit is lost, or confirming receipt before moving on (acknowledgements in networking).

💡 Teaching Tip ▲

If students struggle to suggest solutions, prompt them to think about how online services fix errors (e.g., file downloads that resume from where they left off).

☑️ **Do This:** Define *Transmission Control Protocol (TCP)*.

💡 Teaching Tip ▲

It's okay to briefly introduce TCP here, but there's no need to go into full detail. In an upcoming lesson, students will create their own TCP-like protocol using the Internet Simulator, where they will develop a deeper understanding of how TCP ensures reliable data transmission. This introduction should simply help them recognize that there are rules in place to correct errors and organize data.

🎥 **Video:** Show the second half of the video "The Internet: Wire, Cables, & WiFi" .

🗣️ **Discuss:** *Why might we use one transmission method over another in different situations?*

💡 **Discussion Goal:** Students should recognize that different transmission methods (electricity, light, and radio waves) have trade-offs in speed, range, and reliability. The internet choose transmission methods based on the situation (e.g., WiFi for mobility, fiber optics for speed).

Wrap Up (5 minutes)

☑️ **Do This:** Review the learning objectives.

☑️ **Do This:** Have students take out their The Fabric of the Internet and AI Unit Guide.

☑️ **Do This:** Place students in groups of three or four.

☑️ **Do This:** Direct students to briefly share their thoughts about how messages are broken into bits and sent across networks before individually writing their own response to the Question of the Day in their Unit Guide.

💡 Teaching Tip ▲

Encourage students to build on each other's ideas instead of just repeating the same answer. When writing their individual responses in the Unit Guide, students should include at least one specific example from today's activity.

✓ Assessment Opportunity ▲

The Question of the Day effectively assesses the lesson objectives if students provide responses that explain:

- how binary data is transmitted across a network.
- how network structure and protocols improve the reliability of data transmission.

☑ **Do This:** Display the sentence starters for the Question of the Day for students who may need extra scaffolding for reflection.

💡 Teaching Tip ▲

While all students may not need these sentence starters, displaying them for the class provides optional support without singling anyone out. Encourage students to use a starter if they're unsure how to begin their response, and remind them to include specific examples from the activity.



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Lesson 3: IP Addresses

45 minutes

Overview

Why do devices on the Internet need addresses?

In this lesson, students explore how IP addresses enable communication across networks by acting as unique identifiers for devices. They trace the flow of data through a network, identify issues caused by duplicate IP addresses, and examine how AI optimizes network traffic and reliability.

Standards

Full Course Alignment

CSTA K-12 Computer Science Standards (2017)

- ▶ **CS** - Computing Systems
- ▶ **NI** - Networks & the Internet

CSTA K-12 Computer Science Standards (2026)

- ▶ **SYS** - Systems and Security

Agenda

Warm Up (5 minutes)

Activity (35 minutes)

IP Addressing and Visualizing Networks

Wrap Up (5 minutes)

Objectives

Students will be able to:

- Demonstrate how IP addresses facilitate data transmission and routing
- Explain why unique addresses are necessary for devices to communicate effectively

Preparation

- Print copies of the IP Addressing and Visualizing activity guide (one for each student)
- Print and cut duplicate name cards (using four selected names) so each student receives one; prepare to have students communicate using these during the warm-up.

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the students

- **IP Addressing and Visualizing** - Activity Guide [▼ Make a Copy](#)
- **The Internet: IP Addresses and DNS** - Video

Vocabulary

- **IP address** - a unique number assigned to each device on a network

Teaching Guide

Warm Up (5 minutes)

Warm Up Set Up: To simulate the confusion caused by non-unique identifiers (like repeated IP addresses), students will be given duplicate name cards and attempt to communicate across the room.

- Choose four names (student names or randomly selected ones) and enter them on this slide.
- Print enough copies so each student receives one name card.
- Cut and distribute the cards as students enter the room. Make sure each name is used more than once.
- Select one name to use as the target on the next slide (e.g., “Find out Alex’s favorite food”).

Goal: Set up the confusion caused by non-unique identifiers in a network. Let the confusion and chaos highlight the problem before students are asked to explain it.

 **Distribute:** Hand out name cards as students enter. Some names should be repeated.

☒ **Do This:** Read the prompt aloud and encourage students to move around and ask each other questions to find “[Alex]’s favorite food.”

Teaching Tip

This activity doesn’t need to take more than a minute or so. The goal is to create just enough confusion for students to feel the problem—multiple people sharing the same name—so the connection to IP addresses lands clearly in the follow-up discussion. Keep it fast, light, and playful to build energy at the start of class.

Discuss:

- What problems came up when you started communicating?
- How might you solve these problems?

 **Discussion Goal:** Help students identify challenges like confusion, overlapping answers, and miscommunication. This connects directly to the concept of unique addresses on a network.

☒ **Do This:** Use this slide to transition from student experience to Internet analogy. Read both speech bubbles aloud.

☒ **Do This:** Define IP address as a unique number assigned to each device on a network.

Teaching Tip

Emphasize that just like using someone’s street address or phone number, an IP address helps ensure the message gets to the right place. Ask students if they’ve ever texted the wrong “Alex” and what happened.

- ☑ **Do This:** Introduce the lesson objectives.
- ☑ **Do This:** Introduce the Question of the Day.

Activity (35 minutes)

IP Addressing and Visualizing Networks

- ☑ **Do This:** Place students in pairs.

 **Distribute:** Give each student a copy of the IP Addressing and Visualizing Networks activity guide. Ensure that each pair has access to at least three different colored pens (or pencils, highlighters, etc.) for visualizing AI driven solutions in later scenarios.

- ☑ **Do This:** Instruct students to assign the correct IP addresses to the devices on the Network Diagram of their handout.

 **Circulate:** Check that students are correctly labeling devices based on the routing table. Watch for students mistakenly assigning the same IP to multiple devices or skipping octets.

 **Discuss:** Click on the animated slide to display the prompts:

- *Why do devices on a network need unique IP addresses?*
- *What might happen if two devices had the same IP address? How could this impact communication?*
- *How does the router know where to send data packets when multiple devices are connected to it?*

 **Discussion Goal:** Students should understand why unique IP addresses are essential for clear and reliable communication within a network. Guide students to draw parallels to real-world addressing systems (e.g., postal addresses, phone numbers) and discuss how unique identifiers prevent confusion. Encourage them to consider the consequences of duplicate IP addresses, such as misdirected or lost data, similar to multiple people sharing the same phone number. Additionally, students should connect this concept to the routing table, recognizing how routers use IP addresses to efficiently direct network traffic.

- ☑ **Do This:** Have students take out their The Fabric of the Internet and AI Unit Guide.

- ☑ **Do This:** Define *IP address*.

 **Transition:** Direct students to Lesson 4, Level 1 to complete Levels 1 through 4 to illustrate the flow of information in each level's scenario on their network diagrams by connecting devices, drawing arrows, and using colors for AI solutions.

- **Level 1:** Students illustrate the flow of information in a local network scenario.
- **Level 2:** Students illustrate the flow of information in a scenario with a local network that is connected to an ISP.
- **Level 3:** Students illustrate the flow of information in a scenario involving requesting and receiving data from a website server.
- **Level 4:** Students illustrate the flow of information in three scenarios involving AI-based solutions for network efficiency.

 **Circulate:** As students explore the scenarios, encourage them to actively interact with the network diagram handout by drawing arrows.

 Assessment Opportunity

This activity can serve as an informal formative assessment as the teacher circulates and listens to student discussions. Listen for students explaining routing and IP addressing correctly, check that students are correctly applying lines, labels, and colors, and ask a few students to walk through their network diagrams to ensure they understand the flow of data.

 **Discuss:** Click on the animated slide to display the prompts.

- *How do IP addresses enable communication between devices on different networks, even across the globe?*
- *What are some advantages of using AI to assist with routing compared to traditional methods?*

 **Discussion Goal:** Students recognize that IP addresses allow devices to communicate globally by providing unique identifiers that routers use to forward data. Encourage them to connect these identifiers to real-world identifiers they are already familiar with like address numbers on homes. Students also are able to explain ways in which AI can improve routing efficiency, such as detected congestion, resolving IP conflicts, and adjusting traffic in real time.

Wrap Up (5 minutes)

 **Video:** Show the video "The Internet: IP Addresses & DNS" (0:00–3:23).

 Teaching Tip

Videos are used throughout the curriculum to spark discussions, supplement key concepts with additional explanations and examples, and expose students to the various roles and backgrounds of individuals in computer science.

While interacting with the video, turn on closed captioning so students can also read along as they watch.

To encourage active engagement and reflection, use one or more of the strategies discussed in the [Guide to Curriculum Videos](#).

☒ **Do This:** Review the learning objectives.

☒ **Do This:** Direct students to briefly share their thoughts why devices on the Internet need addresses before individually writing their own response to the Question of the Day in their Unit Guide.

 Teaching Tip

Encourage students to go beyond basic definitions by making connections to today's activities.

 Assessment Opportunity

The Question of the Day effectively assesses the lesson objectives if students provide responses that explain:

- Why unique IP addresses are necessary for devices to communicate reliably

- How routers use IP addresses to direct traffic (at a high level)
- An example from the lesson, such as how AI improves network efficiency



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Lesson 4: Routers and Redundancy

45 minutes

Overview

How does data find its way across the Internet and how does redundancy improve reliability?

This lesson engages students in hands-on exploration of how routers direct data and how redundancy ensures network reliability. Through using an Internet Simulator and diagramming network paths, students analyze router logs and participate in discussions that connect real-world detours to data transmission, deepening their understanding of network resilience.

Standards

Full Course Alignment

CSTA K-12 Computer Science Standards (2017)

- **NI** - Networks & the Internet

CSTA K-12 Computer Science Standards (2026)

- **SYS** - Systems and Security

Agenda

Warm Up (5 minutes)

Activity (35 minutes)

Internet Simulator and Networks Diagram Analysis

Closing

Wrap Up (5 minutes)

Objectives

Students will be able to:

- Analyze how routers help direct data across networks
- Evaluate how redundancy improves the reliability of the Internet

Preparation

- Print copies of the Network Diagram Analysis activity guide (one for each student).

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the students

- **Network Diagram Analysis** - Activity Guide [▼ Make a Copy](#)

Vocabulary

- **redundancy** - having multiple paths so information can still get through if something gets wrong.

Teaching Guide

Warm Up (5 minutes)

 **Discuss:** Click through the animated slide to display the prompts.

- *What happens when a road is closed? How do cars still reach their destination?*

- What would happen if there were no other roads available?

Discussion Goal: Students should recognize that detours keep traffic moving even when there's an issue (redundancy ensures reliability). Without alternative routes, traffic (or data) cannot reach its destination (no redundancy = failure). Guide students to understand that, just like roads, the Internet has many paths for data to travel. If one path is closed or busy, data can take a detour. This redundancy ensures the Internet remains reliable, even during failures or high traffic.

 Teaching Tip

Avoid explicitly mentioning the Internet during the warm-up. Let students connect the analogy to networks as the lesson progresses. The warm-up primes them to understand redundancy by introducing it conceptually without explicitly framing it in technical terms.

If time allows, extend the discussion by asking students for other examples: "What are some other systems where backups or alternatives are important? For example, what happens if your phone dies—how do you stay in touch?"

☒ **Do This:** Introduce the lesson objectives.

☒ **Do This:** Introduce the Question of the Day.

Activity (35 minutes)

Internet Simulator and Networks Diagram Analysis (30 minutes)

Remarks

Redundancy means having backup paths. Just like cars can reroute when there's a roadblock, redundancy on the Internet allows data to reroute when parts of the network have issues.

☒ **Do This:** Define redundancy.

☒ **Do This:** Put students into groups of two.

 **Distribute:** Provide each student with a copy of *Network Diagram Analysis* activity guide.

 Teaching Tip

Consider assigning rotating roles within each pair—Navigator (operates the simulator) and Logger (records diagrams and notes)—to ensure both students stay engaged and contribute meaningfully.

 **Transition:** Direct students to Lesson 4, Level 1 to complete Levels 1 and 2.

- **Level 1:** Students use the Internet Simulator to diagram messages sent through the network.
- **Level 2:** Students complete a check for understanding.

☒ **Do This:** Before students begin to work, use slides 9-15 to review the instructions for the activity in order to demonstrate how to use the Internet Simulator in order to diagram the path their messages take.

☑ **Do This:** Introduce the Internet Simulator and how students should join. They can choose any of the routers as long as it's not the same as their partner's.

💡 Teaching Tip

Add up to as many routers as there are students in your classroom. This will help illustrate how many hops across multiple routers their messages take.

You can also assign multiple students to a single router, just make sure that elbow partners are on different routers.

☑ **Do This:** Explain that students need to know the IP address to send their partner a message.

☑ **Do This:** Explain how to send a message to their partner.

☑ **Do This:** Explain how to navigate to the "Log Browser" from the "Router" tab to access the message data.

☑ **Do This:** Click through the animated slide to explain how to read the logs.

First, choose the "show traffic to me" option in the first dropdown menu and "show all routers" in the second dropdown menu. The path that these messages take are what you will be diagramming. If we look at this message that asks "What's your favorite color?", we see that the message starts at Router 1 because this person has an id of router 1.1, then it goes to router 8, 6, and finally Router 2. This person's partner is at Router 2.12, so the message got to them. If you're unsure which router was pinged first, look at the timestamp in the first column.

☑ **Do This:** Explain that students should diagram the path their message took on their activity guide for up to three messages.

☑ **Do This:** Click through the animated slide to demonstrate adding the nodes and IP addresses of their device and their partner's device.

🎤 Remarks

A node represents any device that connects to a network—this could be a phone, computer, printer, router, or even a smart thermostat. Routers play a crucial role in networking because they connect multiple devices (or nodes) to the Internet and help direct data between them. While routers themselves are also considered nodes, their primary function is to manage network traffic and assign local IP addresses to devices within a network.

🖥️ **Transition:** Direct students to Lesson 5, Level 1 to complete Levels 1 and 2.

- **Level 1:** Students use the Internet Simulator to diagram messages sent through the network.
- **Level 2:** Students complete a check for understanding.

🖥️ 1-2

Internet Simulator and Networks Diagram Analysis

1

2

🔄 **Circulate:** Monitor students as they send messages. While they will diagram all messages sent over the network, ensure they only send messages to their partner. Check that they use the router logs correctly to follow the path each message took and ensure their diagrams accurately represent those paths.

💡 Teaching Tip

Let students know that it is okay if they don't have time to diagram every single message, especially if you have a large number of messages. The goal is for them to diagram enough messages to see that they are not taking direct paths to their destinations, but rather bouncing through multiple routers, and the paths messages are taking will most likely not be the same exact path each time.

✓ Assessment Opportunity ▲

The activity guide can be used as a formative assessment of student understanding of routers and the idea of data taking different paths (redundancy).

☑ **Do This:** Direct students to self-assess their understanding by choosing the answer that best explains what redundancy means for networks and how it helps keep the Internet reliable.

☑ **Do This:** Review the correct answer (B) quickly before moving students into discussing their diagrams.

Closing (5 minutes)

🗣️ **Discuss:** Click through the animated slide to display the prompts.

- *How did the redundancy in your network help prevent complete failure?*
- *What might have happened if no redundant paths were available?*

💡 **Discussion Goal:** Students explain how redundancy in a network helps maintain connectivity by providing alternative paths when a connection fails. They describe specific examples of how redundant paths allow data to continue flowing, such as rerouting traffic through different nodes or links. Additionally, students discuss the potential consequences of having no redundancy, including total network failure, loss of communication, or disruptions in internet access.

🗣️ **Discuss:** Click through the animated slide to display the prompts.

- *Why do you think routers chose new paths for some messages?*
- **How might redundant paths be important during high-traffic events, like streaming live sports or during natural disasters?*

💡 **Discussion Goal:** Listen for and guide students to the following points: Routers often choose new paths for messages to avoid problems like traffic jams or broken connections. If one path is too busy or isn't working well, the router can send the message along a different route so it still reaches its destination quickly and reliably. This process is similar to taking a detour on the way to school when your usual road is closed or too congested. Redundant paths are like backup routes on a map. During high-traffic events, such as live sports streaming when many people are online, or during natural disasters when some parts of the network might be damaged, these extra routes help keep the network working smoothly. If one path gets overloaded or fails, data can still travel through another route, ensuring that information gets to its destination without interruption.

Wrap Up (5 minutes)

☑ **Do This:** Review the learning objectives.

☑ **Do This:** Have students take out their The Fabric of the Internet and AI Unit Guide.

☑ **Do This:** Review the definition of redundancy.

☑ **Do This:** Have students write down idea for how data finds its way across the internet and one idea for how redundancy improves reliability. Instruct students to jot their ideas down in bullet points rather than complete sentences to allow time for them to share their ideas with a Give One, Get One protocol.

☑ **Do This:** Play the music clip to cue the Give One, Get One activity and direct students to participate in a Give One, Get One to share their ideas to the Question of the Day.

💡 Teaching Tip ▲

Using music to cue transitions helps create routines in the classroom. When students hear cues, they know it is time to transition to a specific activity. Emphasize to students what the music clip means, and be consistent with the use of the music clip to cue the activity. Click the blue play icon on the slide to play the music cue.

Give One, Get One encourages students to seek and share ideas and information. Students circulate the classroom and pair up with a partner when prompted by the teacher. Each student "gives" or shares one of their ideas as the other student "gets" or listens and writes it down. The teacher then prompts students to find new partners and repeat the process.



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Lesson 5: Packets

45 minutes

Overview

How does data travel reliably to reach its destination on an unreliable network?

In this lesson, students explore how data is divided into packets for transmission and the challenges of ensuring reliable delivery. Through an unplugged simulation and an interactive Internet Simulator activity, students experience real-world issues like packet loss and delays. They develop and refine protocols to improve reliability. The lesson concludes with a discussion on TCP and UDP in order to highlight the tradeoffs networks face between reliability and speed.

Standards

Full Course Alignment

CSTA K-12 Computer Science Standards (2017)

- ▶ **CS** - Computing Systems
- ▶ **NI** - Networks & the Internet

CSTA K-12 Computer Science Standards (2026)

- ▶ **SYS** - Systems and Security

Agenda

Warm Up (5 minutes)

Activity (35 minutes)

Wrap Up (5 minutes)

Objectives

Students will be able to:

- Demonstrate how data packets are reassembled after transmission
- Explain how data is divided into packets for transmission

Preparation

- Review the Packets Unplugged Teacher Guide and decide:
 - how you will arrange students in the classroom
 - how you will assign IP addresses and partners for each student
- Print copies of the Packets and Protocols activity guide (one for each student)

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the students

- **Packets and Protocols** - Activity Guide [▼ Make a Copy](#)
- **The Internet: Packets, Routing, and Reliability** - Video

Vocabulary

- **User Datagram Protocol (UDP)** - a protocol that focuses on speed and

does not guarantee all packets
arrive

Teaching Guide

Warm Up (5 minutes)

 **Discuss:** Click through the animated slide to display the prompts.

- *Imagine you're sending a big package, but have to break it into smaller boxes. What challenges might you face in getting everything delivered safely?*
- *What might happen to some of the boxes along the way?*
- *What could make sure they all arrive?*

 **Discussion Goal:** This discussion sets the stage for the challenges students will encounter in the unplugged activity. Encourage students to consider challenges like packages getting lost or delayed, arriving out of order, as well as ways to improve reliability, such as labeling and tracking.

Teaching Tip

Do not explicitly mention labeling, lost boxes, or reconstructing the message. Let students arrive at these challenges naturally during the unplugged activity.

☒ **Do This:** Introduce the learning objectives.

☒ **Do This:** Introduce the Question of the Day.

Activity (35 minutes)

 **Distribute:** Give each student a piece of scratch paper.

☒ **Do This:** Direct students to make a simple drawing, or write a handwritten message on their scratch piece of paper.

Remarks

You have just made a message that you are going to send to someone else on the other side of the room. However, we're going to imagine that it's too big to send all at once. So tear it into four pieces.

☒ **Do This:** Assign each student an IP address and a partner.

Teaching Tip

Depending on the context of each classroom, assigning partners and IP addresses may look different. Some possible implementations are below:

- Project a seating chart with pre-assigned IP addresses
- Count students in real time to assign IP addresses

- Provide general guidelines for IP addresses and allow students to choose their own (e.g. "It must be two numbers between 1-100, separated by a period.")

☑ **Do This:** Ask students to look across the room to find a partner - they can point at or make eye contact with someone. (Students in the middle should find partners who aren't at the middle table). Once each student has a partner, ask students to tell their partner their IP address.

☑ **Do This:** Instruct students to address each of their pieces. Students should write their partner's IP address on each piece of paper that makes up their drawing, clearly indicating that it is going to that address. For example - "To: 4.2"

☑ **Do This:** Explain the rules to the students, and check that all students understand what they need to do.

☑ **Do This:** Click through the animated slide to demonstrate paths a message might take, and explain that messages may go missing or arrive out of order.

☑ **Do This:** Call "Go!" to begin the activity!

💡 Teaching Tip

There will probably be enough "traffic" going around for issues of delayed packets and dropped packets to emerge. But if you want to force the issue you can try a few of the following:

- As students are passing paper, teacher can participate or try to interfere by delaying and occasionally destroying pieces of paper.
- You may also prepare a different drawing to send to one student, just to introduce some confusion.

☑ **Do This:** Call "Stop!" after about five minutes, or at least one student was able to receive and reconstruct the image.

💬 **Discuss:** Click through the animated slide to display the prompts.

- *What challenges did you experience in passing your packets?*
- *What happened if packets didn't arrive or were delayed?*
- *How could we address these issues in a protocol?*

💡 **Discussion Goal:** Students discuss a variety of challenges in transmission (lost or delayed packets, etc) and reassembly. Without necessarily defining specific protocols, encourage students to recognize that rules are needed to ensure packets arrive accurately and in order.

💡 Teaching Tip

Try using the Pairs and Squares protocol to facilitate this discussion. Students discuss each prompt with their partner, and then share with a group of four. The teacher calls on one or two groups and asks for a representative to summarize their discussion.

💻 **Transition:** Direct students to Lesson 5, Level 1. Students explore the changes to the Internet Simulator by sending messages to their partner at a different router.

 **Circulate:** As students explore encourage them to point out the ways in which the simulator has changed.

 Teaching Tip

Use a two or three-minute timer to keep exploration focused and to ensure students have time to for the next part of the activity.

 **Discuss:** Click through the animated slide to display the prompts.

- *What did you notice about the Internet Simulator in this level?*
- *What do you wonder about the Internet Simulator in this level?*

 **Discussion Goal:** Students mention new features like packets being limited to 80 bits and a new "Add Packet" button for splitting messages into multiple packets. Encourage students to explore how whether or not they received all of the packets they were expecting, and if they were in the proper order.

☒ **Do This:** Highlight the new features of the Internet Simulator:

- Packets are limited to 80 bits (64 bits for data, 16 bits for metadata).
- A message can be split into multiple packets using the "Add Packet" button.
- Packets may arrive out of order or get dropped during transmission.

 **Distribute:** Give each pair of students a copy of the Packets and Protocols activity guide.

☒ **Do This:** Instruct students to draft a protocol to address packet challenges they experienced during the unplugged activity (e.g., numbering packets, confirming receipt, requesting missing packets).

☒ **Do This:** Instruct students to test their protocols on Level 1.

 **Circulate:** Circulate, observe, and ask guiding questions such as "What happens if your confirmation message gets dropped?" or "How will you handle packets arriving out of order?"

 Teaching Tip

- **Emphasize Iteration and Learning:** "It's okay if your protocol isn't completely perfect! The goal of this activity isn't to design a flawless protocol but to discover the challenges involved in ensuring reliable data transmission. Focus on exploring what issues your protocol needs to address and how it can handle those challenges."
- **Encourage Flexibility:** "If your first protocol doesn't work as expected, that's fine—use it as an opportunity to test and refine your ideas. This mirrors real-world protocol development, which often involves multiple iterations."
- **Reinforce the Key Takeaway:** "The point of this activity is to realize how complex it is to design a protocol that accounts for dropped, delayed, or out-of-order packets. Even if your protocol isn't perfect, you're learning to think like a network engineer!"

☒ **Do This:** Instruct students to work with AI chat to revise their protocols on Level 2.

 **Circulate:** Observe student chats and listen for how they interpret the AI's suggestions. Ask guiding questions like:

- "What part of the AI's idea do you think would work best in your simulation?"
- "Is there anything the AI suggested that wouldn't fit here?"

Teaching Tip

- Encourage students to treat the AI as a collaborator, not the final answer.
- Remind them to explain why they kept, changed, or rejected AI ideas.

☒ **Do This:** Instruct students to test their revised protocol on Level 1 and complete the Check for Understanding on Level 3.

Teaching Tip

Teachers can display anonymous student responses and go over misconceptions or ask students what similarities and differences they notice about the responses

Assessment Opportunity

The check for understanding question can be used to formatively assess student understanding of how data packets are reassembled after transmission.

☒ **Do This:** Remind students of the TCP protocol they learned about in a previous lesson. Guide students to make the connection that they they just created a TCP protocol with their partner.

☒ **Do This:** Introduce the UDP protocol to students.

Video: Show the video "The Internet: Packets, Routing, and Reliability"

Discuss: *What would simulating UDP in the Internet Simulator look like?*

Discussion Goal: Students mention that UPD is focused on speed, so simulating it on the Internet simulator would involve sending packets without checking for errors or missing data.

Teaching Tip

To encourage active engagement and reflection, use one or more of the strategies discussed in the [Guide to Curriculum Videos](#).

Wrap Up (5 minutes)

☒ **Do This:** Review the learning objectives.

☒ **Do This:** Have students take out their The Fabric of the Internet and AI Unit Guide.

☒ **Do This:** Place students in groups of 3-4.

☑ ****Do This: ****Direct students to briefly share their thoughts on how data travels reliably despite an unreliable network before writing their own response to the Question of the Day in their Unit Guide.

💡 Teaching Tip ▲

Encourage students to build on each other's ideas instead of just repeating the same answer. When writing their individual responses in the Unit Guide, students should include at least one specific example from today's activity.

✔ Assessment Opportunity ▲

The Question of the Day effectively assesses the lesson objectives if students provide responses that explain:

- How network protocols (e.g., TCP) ensure reliable data transmission despite dropped, delayed, or out-of-order packets.
- The role of packet numbering, acknowledgments, and retransmission in ensuring messages are reassembled correctly at their destination.



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Lesson 6: HTTP and DNS

45 minutes

Overview

How do websites find their way to your browser using DNS and HTTP?

This lesson explores how DNS (Domain Name System) and HTTP (HyperText Transfer Protocol) work together to connect devices across the Internet. Students engage in hands-on activities, including an IP address collection exercise, a DNS query simulation, and an HTTP request exploration, to understand how devices locate websites and retrieve web content. Through discussions, chatbot interactions, and network monitoring exercises, students analyze how DNS translates domain names into IP addresses and how HTTP requests enable seamless access to online resources.

Standards

Full Course Alignment

CSTA K-12 Computer Science Standards (2017)

- ▶ **CS** - Computing Systems
- ▶ **IC** - Impacts of Computing
- ▶ **NI** - Networks & the Internet

CSTA K-12 Computer Science Standards (2026)

- ▶ **SYS** - Systems and Security

Agenda

Warm Up (5 minutes)

Activity (35 minutes)

Internet Simulator: DNS Queries

The DNS Hierarchy

AI Chat Inquiry

HTTP Preview and Inspect Tool Exploration

Wrap Up (5 minutes)

Objectives

Students will be able to:

- Describe the function of HTTP in requesting web pages
- Explain how DNS resolves domain names into IP addresses

Preparation

- Check that all student devices can open Developer Tools > Network Tab for HTTP observation
- Print and cut out IP Address Labels for the warm-up activity
- Print IP Address Collection Activity Guide (one for each student)
- Print DNS Hierarchy Handout (one for each student)

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the students

- **HTTP Request Simulator** - Resource [▼ Make a Copy](#)
- **DNS Hierarchy & Exploring Energy Costs** - Handout [▼ Make a Copy](#)
- **IP Address Collection** - Activity Guide [▼ Make a Copy](#)
- **Scavenger Hunt Activity** - Handout [▼ Make a Copy](#)

Vocabulary

- **Domain Name System (DNS)** - a system that translates human-friendly website names (like [www.example.com](#)) into numerical IP addresses that computers use to find and communicate with each other on the Internet
- **HyperText Transfer Protocol (HTTP)** - a protocol that allows computers to request and share web content, like webpages, images, and videos

Teaching Guide

Warm Up (5 minutes)

 **Distribute:** As students enter class hand each of them a single IP address. Also give each student a copy of the IP Address Collection activity guide.

 **Do This:** Explain to the students that today we're going to finish our study of the layers of the Internet and learn about two final protocols and we're going to just hop right in! Inform students that when they walked into the room you handed each student a slip of paper - this is their IP address. Remind students that we know that in order to communicate on the internet you need to know the IP address of the other computer and that so far we've let you ask one another's addresses when communicating on the Internet Simulator, but this can be tricky.

 **Do This:** Explain that students' goal is to create a list of every one of your classmates IP addresses. The only rules:

- They may walk around the room
- They may share information with classmates
- They may talk and share information with only ONE classmate at a time

 **Circulate:** As students are working, circulate quietly through the room and do the following:

- Approach a student and silently take their IP address slip away from them.
- Give that person a new IP address slip (or a re-used IP address).
- Repeat the above two steps as many times as you can, as you circulate the room.

Teaching Tip

This quick unplugged activity helps students understand why the DNS exists in the first place. With devices joining the internet all the time it's impossible for each device to keep track of each other device's IP address.

This activity can be done very quickly, typically in about 3 minutes, before students start to see the core challenge. If students start running away from you that's a good sign they've seen what makes this activity tricky!

 **Discuss:** Click through the animated slide to display the discussion prompts.

- *Why do you think I was switching IP addresses?*
- *If IP addresses can change, is there a better way for everyone to know everyone else's IP address?*

 **Discussion Goal:** The main takeaways from the discussion are:

- Devices are joining and leaving the Internet all the time. IP addresses don't stay constant. If you go to a coffee shop or restart your browser your device might be assigned a new IP address.
- If IP addresses are switching, it's very hard for each computer to keep an accurate list.
- It would make more sense if there were one system that kept track of all that information

☒ **Do This:** Introduce DNS to students.

☒ **Do This:** Introduce the learning objectives.

☒ **Do This:** Introduce the Question of the Day.

Activity (35 minutes)

Internet Simulator: DNS Queries (15 minutes)

☒ **Do This:** Explain the goal and rules for the Internet Simulator DNS task and answer any questions they may have before transitioning to the computer. Let students know that their task will allow them to see how DNS solves the problem of having to keep track of IP addresses using GET requests to automatically retrieve these addresses.

 **Transition:** Direct students to Lesson 6, Level 1 to use GET requests to obtain at least two classmates' IP addresses and have a quick conversation.

 1

Internet Simulator: DNS Queries

 **Circulate:** Ensure that students are correctly sending GET requisitions and using retrieved IP addresses.

 Teaching Tip

Students shouldn't need more than 5 minutes in order to gain insight into how DNS uses GET requests to solve the problem of keeping track of IP addresses and resolving human-readable usernames (or domain names) into IP addresses. Once you feel that your students have grasped this concept, you can move on the debrief and discuss DNS.

 **Discuss:** Click through the animated slide to display the discussion prompts.

- *What problem does DNS solve?*
- *What did you notice about how GET requests work to retrieve information?*

 **Discussion Goal:** Listen for answers that point out how DNS converts human-friendly usernames (or domain names) into machine-readable IP addresses and that GET requests structure how information is requested. DNS makes our lives easier by automatically handling the translation of names to IP addresses—just like a phone book.

Video: Show the video “The Internet: IP Addresses & DNS” from 4:10 to the end of the video. Have students focus on how DNS translates domain names and how the system scales.

Teaching Tip

To encourage active engagement and reflection, use one or more of the strategies discussed in the [Guide to Curriculum Videos](#).

The DNS Hierarchy (10 minutes)

Distribute: Give each student a copy of the DNS Hierarchy handout.

Do This: Explain that while the video gave a good summary of how DNS works, the reality of the process has many more layers! Their job is to quickly work with a DNS chatbot to fill in some missing information about the entire DNS Hierarchy and its process.

Transition: Direct students to Lesson 6, Level 2 to chat with the DNS Hierarchy chatbot to help them fill in the missing components in their handout.



DNS Hierarchy Chatbot

Circulate: Ensure that students are using the suggested prompts to ask the chatbot for the missing information.

Teaching Tip

Students shouldn't need more than 5-8 minutes in order to interact with the chatbot to fill in their handout.

Do This: Before students begin the chatbot activity, show the example conversation on the slide to model effective questioning.

Discuss: Click through the animated slide to display the discussion prompts.

- *What problem does DNS solve?*
- *What did you notice about how GET requests work to retrieve information?*

Discussion Goal: Listen for answers that point out how DNS converts human-friendly usernames (or domain names) into machine-readable IP addresses and that GET requests structure how information is requested. DNS makes our lives easier by automatically handling the translation of names to IP addresses—just like a phone book.

Discuss: Click through the animated slide to display the discussion prompts:

- *What do you notice about the DNS hierarchy?*
- *What do you wonder about how it the hierarchy works?*

Discussion Goal: Emphasize to students that this distributed hierarchy ensures scalability and efficiency, allowing DNS to handle billions of requests daily. Facilitate a discussion to clarify roles and highlight key points:

- Recursive Resolver starts the process and queries other servers.
- Root Servers guide the request to the correct TLD server.
- TLD Servers handle requests for .com, .org, etc., and direct them to the Authoritative Server.

- Authoritative Servers provide the IP address.

AI Chat Inquiry

☒ **Do This:** Direct students to Lesson 6, Level 4 to chat with the AI about the energy costs of Internet and AI infrastructure.



3

AI Chat: Energy Costs of the Internet

 **Circulate:** Check that students are using the suggested prompts to guide their conversation with the AI and recording their responses on their activity guides. Encourage them to ask follow-up questions to clarify or verify the AI's explanations.

Teaching Tip

Students shouldn't need more than 7–10 minutes to complete this AI Chat activity. Emphasize that the goal is to connect what they've learned about HTTP and DNS to the real-world energy impact of online communication and AI systems.

HTTP Preview and Inspect Tool Exploration (10 minutes)

 **Video:** Show the video segment of “The Internet: HTTP & HTML” Have students focus on how browsers retrieve content from servers and what GET and POST requests are.

☒ **Do This:** Click through the animated slide to demonstrate how DNS and HTTP work together to bring a website to a user's browser.

☒ **Do This:** Introduce the HTTP protocol to students.

☒ **Do This:** Click through the animated slide to reinforce the analogy between sending a letter and retrieving a web page.

 **Transition:** Direct students to Lesson 6, Level 3 to complete Levels 4 through 6.

- **Level 4:** Students access the HTTP Cats exploration external site.
- **Level 5:** Students complete a check for understanding.
- **Level 6:** Students complete a check for understanding.

☒ **Do This:** Place students in pairs.

 **Distribute:** Have each pair make a copy of the Scavenger Hunt slides.

 4-6

HTTP Requests Exploration

4

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6

Teaching Tip

If students have access to real Developer Tools, encourage them to complete the scavenger hunt using their own browser, and the files that load on an actual webpage. This gives them hands-on experience with live HTTP traffic and helps reinforce the connection between what they see in the simulator and how browsers work in the real world.

☑ **Do This:** Have students work through the HTTP Scavenger Hunt. They should use either the HTTP Explorer simulator or their own browser to complete each task and fill in the Method, Status Code, File Type, and Load Time fields in their activity guide.

🖥 **Transition:** Direct students to Lesson 6, Level 4 to answer the questions.

✔ Assessment Opportunity ▲

The responses on Levels 5 and 6 can be used to formatively assess the lesson objectives:

- Describe the function of HTTP in requesting web pages and resources

Wrap Up (5 minutes)

☑ **Do This:** Review the learning objectives.

☑ **Do This:** Have students take out their The Fabric of the Internet and AI Unit Guide.

☑ **Do This:** Give students a quick minute to jot down their thoughts on the Question of the Day. Encourage students to include at least one specific example from any of today's activities.

☑ **Do This:** Place students in groups of three or four.

☑ **Do This:** Direct students to briefly share in one or two sentences how websites find their way to your browser using DNS and HTTP.

🔄 **Circulate:** Listen to small-group conversations and if time permits, call on two or three groups to share their key insights with the class.

💡 Teaching Tip ▲

Encourage students to build on each other's ideas instead of just repeating the same answer.

✔ Assessment Opportunity ▲

The Question of the Day effectively assesses the lesson objectives if students provide responses that explain:

- how DNS resolves domain names into IP addresses.
- the function of HTTP in requesting web pages.

🗣 **Discuss:** *How do websites use DNS and HTTP to load in your browser?*

💡 **Discussion Goal:** Listen for answers that synthesize the process: DNS translates domain names into IP addresses, and HTTP uses GET requests to retrieve content from servers, working together to bring websites to life.



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Lesson 7: The Internet's Hidden Barriers

45 minutes

Overview

Who has access to the Internet and who is left behind?

This lesson explores the hidden barriers to Internet access, focusing on the digital divide and net neutrality. Students analyze real-world data from Pew Research to identify disparities in Internet access and engage in a case study simulation where they interact with personas facing digital accessibility challenges. Through discussions, data interpretation, and chatbot interactions, students connect these issues to education, employment, and civic participation, considering solutions to promote equitable Internet access.

Standards

Full Course Alignment

CSTA K-12 Computer Science Standards (2017)

- ▶ **IC** - Impacts of Computing

CSTA K-12 Computer Science Standards (2026)

- ▶ **SOC** - Computing and Society
- ▶ **SYS** - Systems and Security

Agenda

Warm Up (10 minutes)

Activity (30 minutes)

Digital Divide Data Analysis

Case Studies with AI Chatbot

Wrap Up (5 minutes)

Objectives

Students will be able to:

- Evaluate the impact of the digital divide on different populations
- Explain the impacts of net neutrality
- Identify strategies to address challenges to equitable Internet access

Preparation

- Print copies of the Internet Access IRL activity guide.

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the students

- **Internet Access IRL** - Activity Guide

▼ Make a Copy

Vocabulary

- **Net Neutrality** - the idea that all websites and online services should be treated equally by Internet providers
- **digital divide** - the gap between people who have easy access to the Internet and technology and those who do not, often due to differences in income, location, or resources

Teaching Guide

Warm Up (10 minutes)

☒ **Do This:** Explain the Four Corners upcoming activity to students. Make sure to point out which corner will be **Yes, No, Maybe, and Undecided**.

Teaching Tip

If your classroom can't accommodate movement or you have students uncomfortable sharing in large groups, consider using "Line Debate" (students hold up fingers to show agreement level) or silent discussion protocols like "Chalk Talk" or digital polls.

☒ **Do This:** Read the first scenario to students and direct them to move to the corner that best aligns whether they agree, disagree, think maybe or are unsure. Instruct students to discuss their reasoning with others who share their stance and then call on a student from each corner to share why their group chose their position.

Teaching Tip

Encourage critical thinking about fairness, competition, and access on the Internet as well as cost barriers, freedom of information, or business interests.

☒ **Do This:** Read the second scenario to students and direct them to move to the corner that best aligns whether they agree, disagree, think maybe or are unsure. Instruct students to discuss their reasoning with others who share their stance and then call on a student from each corner who was not a spokesperson for scenario #1 to share why their group chose their position.

☒ **Do This:** Read the third scenario to students and direct them to move to the corner that best aligns whether they agree, disagree, think maybe or are unsure. Instruct students to discuss their reasoning with others who share their stance and then call on a student from each corner who was not a spokesperson for the previous two scenarios to share why their group chose their position.

☒ **Do This:** Introduce Net Neutrality to students and that all of the scenarios relate to net neutrality. Explain to students that this means that without net neutrality, Internet Service Providers (ISPs) can prioritize, slow down or even block websites and companies can pay for faster access creating advantages for some and barriers for others.

Teaching Tip

Net neutrality impacts who has access to fast, reliable Internet and who is left behind. If an ISP charges extra for faster speeds, who might struggle to afford it? How could this deepen the digital divide? Encourage students to think about these questions throughout the lesson.

☒ **Do This:** Click through the animated slide to make direct connections between each scenario and net neutrality.

☒ **Do This:** Introduce the lesson objectives.

☒ **Do This:** Introduce the Question of the Day.

Activity (30 minutes)

Digital Divide Data Analysis (15 minutes)

☒ **Do This:** Put students in groups of two. Make sure students have their Unit Guides out and turned to Lesson 7 to record their observations in the next activity.

 **Transition:** Direct students to Lesson 7, Level 1 to complete Levels 1 through 5 to investigate the data from the Pew Research Center about Internet Access and discuss the questions with their partner.

- **Level 1:** Students explore internet use over time.
- **Level 2:** Students explore home broadband use by income.
- **Level 3:** Students explore internet use by community type.
- **Level 4:** Students explore internet use by age.
- **Level 5:** Students explore smartphone dependency by education.



1-5

Pew Research Center: Internet Access

1

2

3

4

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☒ **Do This:** Instruct students to record their observations in their Unit Guides as they discuss the questions on each level.

Remarks

Now that we understand net neutrality, let's explore the bigger picture: Who actually has access to the Internet? Together, we will look at the data from Pew Research and think about who benefits most and who gets left behind. I will put the graphs up on the board. You will also have a version of these graphs on your own computer so you can see them more clearly.



Teaching Tip



In each level, students will have the opportunity to look at the interactive version of each graph by clicking on the link within each level. If time allows, encourage students to open the link in a new tab and use the "find" tool on their browser to quickly find the graph on the report page. Hovering over each data point will give them more information about the graph.

☒ **Do This:** Click through the animated slide to display possible answers to the annotated graph and discuss what students notice and are surprised by.

 **Discuss:** Click through the animated slide to display the prompts.

- *What do you notice about Internet access across different groups (e.g., income, geography, age)?*
- *How might these disparities impact education, employment, or civic participation?*

 **Discussion Goal:** Students should recognize that Internet access is not distributed equally and that disparities exist based on income, geography, education, and age. Encourage students to connect these inequalities to opportunities in education, employment, and civic engagement, considering how limited access can reinforce existing societal divides.

☒ **Do This:** Have students take out their The Fabric of the Internet and AI Unit Guide.

☑ **Do This:** Introduce the digital divide to students.

Case Studies with AI Chatbot (15 minutes)

📄 **Distribute:** Give each pair of students a copy of the Internet Access IRL activity guide.

🖥️ **Transition:** Direct students to Level 6 to choose one of the AI Chatbot personas that has been affected by the digital divide.



Choose a Persona

🔄 **Circulate:** Listen for student discussions about access barriers (e.g., cost, availability, slow speeds). Prompt students to think whether the persona's experience compares to their own.

💬 **Discuss:** Click through the animated slide to display the prompts.

- *What did you learn about your persona?*
- *How does this connect to the disparities we saw in the Pew Research data?*

💡 **Discussion Goal:** The goal of this discussion is to help students connect real-world technology access disparities to the personas they explored and the data from Pew Research. Students should recognize how limited access to the Internet affects different aspects of life, including education, jobs, healthcare, and civic participation. Finally, encourage them to think critically about possible solutions, including the role of policies like net neutrality in shaping digital equity.

Wrap Up (5 minutes)

☑ **Do This:** Review the lesson objectives.

☑ **Do This:** Direct students to write a short response to the Question of the Day and then have student pair up to share responses.

✔ Assessment Opportunity ▲

The Question of the Day effectively assesses the lesson objectives if students provide responses that explain:

- the impact of the digital divide on different populations and potential solutions to address it.
- Net neutrality and digital divide issues shape opportunities in education, jobs, and civic life.



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Lesson 8: Internet Access Advocacy Campaign Project

135 minutes

Overview

Career Journeys

This project-based lesson guides students in developing an advocacy campaign addressing digital equity, the digital divide, and net neutrality. Students explore real-world challenges, analyze case studies, and propose actionable solutions using AI-driven technology and policy recommendations. Through research, collaboration, and peer feedback, students refine their message and create a compelling advocacy campaign.

Standards

Full Course Alignment

CSTA K-12 Computer Science Standards (2017)

- **IC** - Impacts of Computing

CSTA K-12 Computer Science Standards (2026)

- **SOC** - Computing and Society
- **SYS** - Systems and Security

Agenda

Day 1 (45 minutes)

Warm Up

Activity

Wrap Up

Day 2 (45 minutes)

Warm Up

Activity

Wrap Up

Day 3 (45 minutes)

Warm Up

Activity

Wrap Up

Objectives

Students will be able to:

- Explain the role of routers, switches, and servers in ensuring the Internet is reliable and can grow
- Propose AI-driven solutions to improve Internet access and address challenges like the digital divide
- Propose actionable solutions to address key societal impacts of the digital divide and net neutrality

Preparation

- Print copies of the Persona Fact Sheets for each station
- Print copies of the Project Planning Guide (one for each group)
- Print copies of the Gallery Walk Info Tracker (one for each student)
- Print copies of the Peer Feedback Cards for day 3 (2-4 cards for each student)
- Consider providing poster boards, markers, and art supplies (for physical campaigns)

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the students

- **Gallery Walk Info Tracker** – Activity Guide [▼ Make a Copy](#)
- **Internet Access Advocacy Campaign Project Planning Guide** – Activity Guide [▼ Make a Copy](#)
- **Peer Feedback Cards** – Activity Guide [▼ Make a Copy](#)
- **Persona Summary Sheet** – Handout [▼ Make a Copy](#)

Teaching Guide

Day 1 (45 minutes)

Warm Up (5 minutes)

🎥 **Video:** Show the video “Qualcomm Engineers are Improving Safety and Livelihoods for Fishermen in India”.

💬 **Discuss:** *What challenges did the fishermen face without Internet access, and how do similar barriers affect other communities? How did technology help them, and how could AI improve Internet access for others?*

💡 **Discussion Goals:** Students should discuss the barriers to Internet access and recognize that limited connectivity impacts more than just convenience—it affects safety, economic stability, and access to resources. Guide students to identify the specific challenges faced by the fishermen in the video and consider how similar barriers exist in other communities. Encourage students to think about how technology, including AI, can play a role in solving these access issues, setting up connections to their advocacy project.

Activity (35 minutes)

☑ **Do This:** Place students into groups of 2-3.

📄 **Distribute:** Give each group a copy of the Internet Access Advocacy Campaign Project Planning Guide. This will guide them through all three days of the project.

☑ **Do This:** Explain to students that for the next three days, they will work as Internet Access Advocates. This means they will explore real-world challenges related to equitable Internet access, the digital divide, and net neutrality. They will analyze case studies, assess the societal and ethical impacts of these issues, and develop persuasive advocacy campaigns with actionable and AI-driven solutions. Along the way, they will refine their messages through peer review and collaborative discussions.

💡 Teaching Tip

Depending on your class schedule and student pacing, this project may take 3 to 5 class periods to complete. If you're aiming to complete it in 3 days, consider these in-class adjustments:

- Use timers to keep groups on track during Gallery Walk and Design Sprint
- Offer 2–3 focused campaign format options to streamline production
- Keep the pitch draft informal (bullet points, not full scripts) during peer feedback

☑ **Do This:** Click through the animated slide to explain the first part of the project requirements and answer any questions.

☑ **Do This:** Click through the animated slide to explain the rest of the project requirements and answer any questions.

☑ **Do This:** Direct students to read the project rubric and hold on to any questions they may have.

🗣️ **Discuss:** Click through the animated slide to display the prompts.*What do you notice about the project rubric?

- *What does success look like for this project?*
- *What do you notice about the project rubric?*

💡 **Discussion Goal:** Students share what they notice about the project requirements and rubric and identify the tasks they will need to complete for this project.

🗣️ **Discuss:** *What barriers do you anticipate to achieving success on this project?*

💡 **Discussion Goal:** Students identify barriers they anticipate and share any questions they have about the project.

☑ **Do This:** Click through the animated slide to introduce the steps students will take on Day 1.

☑ **Do This:** Click through the animated slide to introduce the steps students will take on Day 2.

☑ **Do This:** Click through the animated slide to introduce the steps students will take on Day 3.

📄 **Distribute:** Give each student a copy of the Gallery Walk Info Tracker.

☑ **Do This:** Place students into groups of three or four. These groups should be different than who they are working with on the project.

☑ **Do This:** Go over the activity instructions with students. Explain that their group will visit each station, read the scenario, and record key information on their project planning guide. Emphasize the importance of focusing on examples, evidence, and key issues for their advocacy campaign.

☑ **Do This:** Assign each corner a persona (Diego, Lisa, Mr. Johnson, Aisha) and direct groups to visit stations, read scenarios, and record key issues, evidence, and examples on their project guide.

🔄 **Circulate:** As students explore the personas, ask: ‘How does this individual’s experience reflect a larger issue with Internet access? What groups face similar barriers?’ If students are only summarizing without analyzing, prompt them to think about solutions: ‘What policies or technologies could improve this person’s access to the Internet?’

💡 Teaching Tip

Adjust structure as needed—assign starting stations, use a timer for rotations, or provide a digital version with links to each station.

☑ **Do This:** Students should move into their project groups and prepare to discuss insights from the gallery walk. Assign or review team roles (Manager, Summarizer, Recorder, Time Keeper). They will use their notes to identify key issues, examples, and target groups for their advocacy campaign.

☑ **Do This:** Direct students to debrief their Gallery Walks with their group. Ensure group Recorders are recording responses in the group’s Project Planning Guide.

Circulate: As groups are discussing, use probing questions to support groups who may be stuck (e.g., “How does this evidence connect to your focus?”). Check shared notes and provide feedback to strengthen their ideas. Keep groups on track by reminding Time Keepers of the schedule.

Do This: Click through the animated slides to discuss different types of advocacy campaigns with students. Briefly explain each format (e.g., social media video, PSA, infographic) and ask students to consider which format would best suit their message and audience. Encourage them to think about clarity, reach, and impact when selecting a campaign type.

Do This: Direct students to turn to the Design Sprint page in their Advocacy Project Planning Guide. Instruct them to discuss each prompt as a group and record a common answer. Remind them to use team roles (Manager, Summarizer, Recorder, Time Keeper) to stay organized and ensure everyone contributes.

Circulate: Encourage students to move beyond identifying problems and focus on action. Ask: “What specific change do you want to see? How will your campaign convince others to care about this issue?” If students struggle with persuasion, suggest they include a real-world example or statistic to strengthen their argument.

Teaching Tip

If students need more support, consider distributing a Persona Summary Sheet to each group. This quick-reference overview can help students connect real-world challenges to their campaign solutions.

Wrap Up (10 minutes)

Discuss: *What stood out to you the most today? Why does this issue matter? What can we do about it?*

- Use the Three W's protocol to facilitate this discussion. First, students discuss answers in pairs or small groups. Then, select three or four students to share a takeaway.

Discussion Goal: Students should reflect on key takeaways from the lesson and connect the digital divide to real-world impacts. Guide them to recognize who is affected, why it matters, and how advocacy can drive change. This discussion will set the stage for their advocacy campaigns by helping them consider how to make their message compelling and actionable.

Day 2 (45 minutes)

Warm Up (5 minutes)

Discuss: Click through the animated slide to display the prompts.****

- *What key issue did your group decide to focus on?*
- *What evidence or examples did you identify to support your campaign?*
- *What is one challenge or question your group still needs to address?*

Discussion Goal: Students should articulate their campaign's key message and analyze whether their current plan effectively supports it. Encourage them to identify missing evidence or areas where their message could be stronger, ensuring their campaign is clear, compelling, and action-oriented.

Circulate: Ensure students are making direct connections between their focus and evidence. Ask guiding questions: “How does your campaign address a real-world problem?” and “What additional information would strengthen your argument?”

Activity (35 minutes)

☒ **Do This:** Click through the animated slide to remind students of the steps students will take today.

☒ **Do This:** Guide students through **Narrowing Your Focus** in the project guide (page 5). Remind them to clarify their focus, audience, evidence, and solutions before drafting their campaign message. Display the guiding questions to keep them on track.

☒ **Circulate:** Check for clear, concise messaging and ask guiding questions like, “What is your key message?” and “Is your solution realistic and actionable?” Encourage refinement before they move to AI Chat.

Teaching Tip

Students will upload their responses to AI Chat later for targeted feedback. Recommend completing this section digitally if possible; if not, have them take a clear photo to upload.

Optionally, rotate group roles so the Timekeeper helps manage pacing (≈7 minutes).

☒ **Do This:** Direct students to Lesson 8, Level 1. They'll upload their focus answers and use AI Chat to test their campaign message. Remind them to record AI feedback and revisions on page 5.

☒ **Circulate:** Ask: “What parts of the AI’s draft work well?” and “How could it sound more like your group’s voice?”



1

AI Message Testing

Teaching Tip

If working on paper, have groups take a clear photo to upload.

☒ **Do This:** Display part 1 of the example pitch and ask for volunteers to read it aloud or have students read it silently.

Teaching Tip: If you suspect you will be short on time, you might choose to provide printed copies of these example pitch slides to groups as a reference while drafting their own pitch.

☒ **Do This:** Display part 2 of the example pitch and ask for volunteers to read it aloud or have students read it silently.

☒ **Do This:** Display part 3 of the example pitch and ask for volunteers to read it aloud or have students read it silently.

☒ **Do This:** Instruct students to use their reflections from AI Message Testing to draft a 1-minute campaign pitch. They can use bullet points, visuals, or paragraphs to organize their ideas clearly.

☒ **Circulate:** Check that pitches are clear, persuasive, and audience-focused.

Teaching Tip

Set a 7-minute timer to keep groups on pace. Remind students this is a working draft, and to focus on clarity and impact, not perfection.

☑ **Do This:** Instruct students to gather peer feedback using the “One Stay, Two Stray” protocol. Use the slide to explain the procedure to students.

🔄 **Circulate:** Encourage students to give specific feedback by asking: ‘What was strong about this campaign? What could be improved?’ If a campaign lacks a clear call to action, prompt reviewers to ask: ‘What specific action does this campaign want people to take?’

💡 Teaching Tip ▲

- Depending on group size, you may want to have two students stay—one to present and one to take notes—while the others rotate.
- Depending on classroom structure, you may want to assign group numbers before the activity to ensure students rotate in sequence (e.g., Group 1 moves to Group 2, Group 2 moves to Group 3, etc.) for a smoother transition.

☑ **Do This:** Use the sample peer feedback responses to model effective peer feedback and clarify any confusion students might have.

☑ **Do This:** Use the sample peer feedback responses to model effective peer feedback and clarify any confusion students might have.

☑ **Do This:** Use the sample peer feedback responses to model effective peer feedback and clarify any confusion students might have.

☑ **Do This:** Direct students to review their Peer Review Feedback Checklist and discuss which feedback was most helpful. Have them decide what revisions will improve clarity, persuasiveness, and feasibility, then make adjustments to their campaign. Remind them they will have 20 minutes next class for final touches before presenting.

🔄 **Circulate:** Encourage students to prioritize impactful revisions and refine their messaging. If a group is struggling, ask guiding questions like, “What change would make your argument stronger?” or “How can you make your message clearer for your audience?”

Wrap Up (5 minutes)

🗣️ **Discuss:** *What made your campaign compelling and actionable? What could be improved to make it even more effective?*

💡 **Discussion Goal:** Students recognize strengths and areas for improvement in their campaigns while connecting their work to real-world advocacy strategies.

Day 3 (45 minutes)

Warm Up (5 minutes)

🗣️ **Discuss:** Click through the animated slides to display the prompts.

- *What is one aspect of your advocacy campaign that you are most confident about so far?*
- *What is one part of your campaign that could still be clarified, improved, or polished?*

💡 Discussion Goal: Students build confidence in their campaigns by highlighting strengths and what is working well so far. They also have an opportunity to think critically about their campaign's weaker points in order to develop a plan for improvement.

Activity (30 minutes)

☑ Do This: Direct students to review their campaign using the key criteria (clear argument, strong evidence, actionable solutions). Remind them to check and refine their materials so they are presentation-ready.

🔄 Circulate: Ensure students are making intentional revisions rather than just small tweaks. Ask guiding questions like, "How does your campaign persuade your audience?" and "Are your solutions realistic and well-supported?" Provide quick feedback on clarity and presentation.

💡 Teaching Tip

Consider setting a timer to help students stay focused. You may want to give a five-minute warning before the final check-in so they can wrap up any last details.

📄 Distribute: Give each student several copies of the Peer Feedback Cards.

☑ Do This: Instruct groups to pair up with another group. Group A presents while Group B listens and records feedback. After each group has presented, they will pair up with another group and repeat the process.

🔄 Circulate: Ensure presenters stay focused and listeners engage with feedback. Encourage students to ask questions and provide specific, constructive feedback. Support groups by clarifying ideas and keeping discussions on track.

Wrap Up (10 minutes)

☑ Do This: Direct students to complete the End of Project Reflection in their Project Planning Guide as well as on Code Studio, Level 1. If time allows, you can have students share their responses with a partner or do a whole class share-out.

✔ Assessment Opportunity

The Project Planning Guide and group's final format of their campaign pitch can be collected and used to formatively assess the lesson objectives:

- Explain the role of routers, switches, and servers in ensuring the Internet is reliable and can grow.
- Propose actionable solutions to address key societal impacts of the digital divide and net neutrality
- Propose AI-driven solutions to improve Internet access and address challenges like the digital divide.

☰ 2-3

Final Project Reflection

2

3



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Lesson 9: Network Security

45 minutes

Overview

Career Journeys

What vulnerabilities exist in networks, and what strategies can mitigate their impact?

In this lesson, students take on the role of cybersecurity analysts to investigate real-world network security threats and identify effective defenses. Through an interactive simulation, they explore vulnerabilities such as phishing, weak passwords, default credentials, and social engineering. After investigating each vulnerability, students propose a mitigation strategy that can be applied in a real-world context.

Standards

Full Course Alignment

CSTA K-12 Computer Science Standards (2017)

- **IC** - Impacts of Computing

CSTA K-12 Computer Science Standards (2026)

- **SYS** - Systems and Security

Agenda

Warm Up (10 minutes)

Activity (30 minutes)

Network Defense

Wrap Up (5 minutes)

Objectives

Students will be able to:

- Identify common network security vulnerabilities
- Propose solutions to mitigate security risks

Preparation

- Preview Level 1 to familiarize yourself with the scenarios, decision points, and key takeaways
- Label the four corners of the room: Phishing, Weak Passwords, Social Engineering, Default Credentials for the lesson wrap up.

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the students

- **Fabric of the Internet and AI Unit Guide** - Activity Guide
[▼ Make a Copy](#)
- **How Cyber Attacks Happen in Real Life** - Video
- **Identify, Defend, Reflect** - Resource
[▼ Make a Copy](#)

Teaching Guide

Warm Up (10 minutes)

🎥 **Video:** Show the video "How Cyber Attacks Happen in Real Life".

💡 Teaching Tip

To encourage active engagement and reflection, use one or more of the strategies discussed in the [Guide to Curriculum Videos](#).

💬 **Discuss:** Click through the animated slide to display the prompts.

- *What vulnerabilities allowed the hacker to breach Acme Corporation's network?*
- *What could Acme have done to prevent the attack?*
- *What impact did the attack have on Acme, and how do you think it could impact their customers?*

💡 **Discussion Goal:** Students recognize that multiple security failures—including weak passwords, social engineering, and multi-factor authentication (MFA) fatigue—enabled the hacker to breach Acme Corporation's network. Guide students to identify how better security practices, such as stronger password policies, improved multi-factor authentication, and offline backups, could have prevented the attack. Encourage students to consider the broader consequences, including financial loss, reputational damage, and compromised customer data.

☑ **Do This:** Introduce the lesson objectives.

☑ **Do This:** Introduce the Question of the Day.

Activity (30 minutes)

Network Defense

☑ **Do This:** Place students in pairs or small groups of three. Make sure students have their Unit Guide out and opened to Lesson 9.

💡 Teaching Tip

The Identify, Defend, Reflect! activity is an interactive game in which students will use a clickable slideshow that will navigate them through various network vulnerabilities. The goals of the activity are to:

- Discover common network vulnerabilities
- Investigate key details of those vulnerabilities
- Identify appropriate defenses in network vulnerability scenarios

The game covers four common network vulnerabilities. Each should take about 7 minutes to complete. We recommend modeling the first vulnerability to demonstrate how to interact with the slideshow.

☑ **Do This:** Introduce the four key security threats students will encounter in the simulation.

☑ **Do This:** Walk students through the sample notes. Point out what kind of info goes in each column, emphasizing quick notes and ideas they might want to return to later.

Remarks

Today, you'll step into the role of a cybersecurity analyst, investigating real-world security threats and deciding how to defend against them. Follow the instructions in the interactive slide deck, record your observations in your Unit Guide, and think critically about how to protect networks from cyber attacks

 **Transition:** Direct students to Lesson 9, Level 1. Lead the class through a demonstration of the first “round” of the interactive slide show. Walk them through one scenario and model how to navigate through the slides and record notes in the unit guide. Then, release students to work with their partners to complete the remaining scenarios.

 1

Identify, Defend, Reflect!

 **Circulate:** As students work through the slides, check that students are following the instructions and recording observations. Listen for discussions between partners—are they just clicking through, or are they analyzing the scenarios. When students are selecting defenses, ask probing questions about what made them choose certain options.

 **Discuss:** *If you were a cybersecurity analyst for Cybershield Inc., which mitigation strategy would you prioritize implementing first, and why?*

 **Discussion Goal:** Students think critically about the mitigation strategies they encountered in the interactive slideshow. They discuss the importance of a variety of mitigation strategies and the consequences of not having those strategies in place.

 **Transition:** Direct students to Lesson 9, Level 2 to complete the Check for Understanding.

 2

Check for Understanding

Wrap Up (5 minutes)

☑ **Do This:** Review the lesson objectives.

☑ **Do This:** Give students a quick minute to jot down their thoughts on the Question of the Day. Encourage students to include at least one specific example from any of today's activities.

Assessment Opportunity

The Question of the Day effectively assesses the lesson objectives if students provide responses that explain:

- common network security vulnerabilities.
- solutions to mitigate security risks.

☑ **Do This:** Label the four corners of the room: Phishing, Weak Passwords, Social Engineering, Default Credentials. Instruct students to go to the corner that relates to the one vulnerability they mentioned in their question of the day answer. Have students share their answers and compare mitigation strategies. If time allows, have one person from each group share one takeaway from their discussion with the class.

☰ 3

Student Survey



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Lesson 10: Emerging Technologies

45 minutes

Overview

Career Journeys

What makes AI effective at enhancing network security, and how does it compare to traditional methods?

This lesson explores how AI enhances cybersecurity by detecting and responding to threats in real time. Through a hands-on simulation and discussions, students analyze how AI compares to traditional security methods and consider the challenges of implementing AI-driven solutions. The lesson encourages students to think critically about AI's benefits, limitations, and role in network security.

Standards

Full Course Alignment

CSTA K-12 Computer Science Standards (2017)

- ▶ **AP** - Algorithms & Programming
- ▶ **NI** - Networks & the Internet

CSTA K-12 Computer Science Standards (2026)

- ▶ **SOC** - Computing and Society
- ▶ **SYS** - Systems and Security

Agenda

Warm Up (5 minutes)

Activity (35 minutes)

AI Threat Detection

Wrap Up (5 minutes)

Objectives

Students will be able to:

- Analyze the role of AI in real-time threat detection
- Propose AI-based solutions to network security challenges

Preparation

- Print and cut out Threat Detected! Cards (one set for each pair of students)
- Print AI Threat Detection Notetaker Activity Guide (one for each pair of students)

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the students

- **AI-Based Real-Time Threat Detection** - Video
- **Threat Detection Guided Notes** - Activity Guide [▼ Make a Copy](#)

Teaching Guide

Warm Up (5 minutes)

 **Video:** Show the video “AI-Based, Real-Time Threat Detection” to students.

 **Discuss:** *How does detecting threats in real time improve network security compared to delayed detection? What challenges might companies face when adopting AI-driven security tools like Morpheus? How could these challenges be addressed?*

 **Discussion Goal:** This discussion prepares students to think critically about the role of AI in modern cybersecurity and its real-world applications. Students analyze how real-time threat detection improves network security, contrasting AI-driven tools with traditional methods that often detect threats too late. Encourage students to consider how faster detection helps prevent breaches and limits damage. Guide them to recognize the challenges organizations face when implementing AI security tools, such as cost, training, or false positives, and how these barriers can be addressed.

 Teaching Tip

Ensure students are making connections between AI's strengths and challenges. Ask questions like, *"What happens if AI misidentifies a threat?"* or *"Why might some organizations be hesitant to rely on AI security tools?"*

☒ **Do This:** Explain to students that today, they are diving into one of the most cutting-edge topics in cybersecurity—how AI is revolutionizing network security. Every second, massive amounts of data are moving through networks, and cyber threats are evolving just as quickly. Traditional security methods struggle to keep up, but AI is changing the game by detecting threats in real time before they cause damage. As they explore AI-driven security tools today, encourage students to think critically about whether AI is the ultimate solution, or does it have its own risks and challenges.

☒ **Do This:** Introduce the lesson objectives.

☒ **Do This:** Introduce the Question of the Day.

Activity (35 minutes)

AI Threat Detection

☒ **Do This:** Put students into groups of two.

 **Distribute:** Give each pair of students a copy of the Threat Detected! Cards and the AI Threat Detection Notetaker activity guide.

 Teaching Tip

For early finishers or advanced groups, provide a set of Tier Two Threat Detected! Cards. These cards introduce more complex or ambiguous scenarios that require deeper analysis of AI limitations, ethical concerns, and the need for human oversight. Encourage students to use the same notetaker and apply their learning from earlier rounds.

Remarks

Today, you will act as security analysts, using an AI threat detection system to investigate and respond to security threats. Your goal is to evaluate how AI-based security tools work compared to traditional detection methods. Work with your partner to complete this activity. You will receive one set of AI Threat Detection Cards and you will each have your own Threat Detection Notetaker to fill out individually.

☑ **Do This:** Explain that students will act as security analysts interacting with an AI threat detection system to analyze and respond to network vulnerabilities. Review the steps for drawing a Threat Detected card, discussing the scenario with the AI Chatbot, recording the results of the discussion on the Activity Guide, and then drawing a new card to repeat the process.

💻 **Transition:** Direct students to Lesson 10, Level 1 to interact with the threat detection chatbot and complete their activity guide.

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Threat Detection Chatbot

☑ **Do This:** Give students at least 20 minutes to complete as many threat cards as possible in that time. Then, direct student to move on to Level 2 to complete a check for understanding.

🔄 **Circulate:** Circulate and listen for students who assume AI is always accurate. If students trust the AI's responses without question, ask: 'Could AI ever misidentify a threat? How would that impact security?' Ensure students also consider limitations like false positives, AI bias, and the need for human oversight.

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Check for Understanding

💡 Teaching Tip

If time permits, display anonymous student responses for the check for understanding on Level 2 and go over misconceptions or ask students what similarities and differences they notice about the responses.

✅ Assessment Opportunity

Level 2 can be used to formatively assess the learning objectives.

☑ **Do This:** Show the model threat detection notes, emphasizing short, clear notes instead of full sentences. Remind students that they don't need to capture every word from the chatbot, just the gist of its response.

☑ **Do This:** Put students into groups of four.

☑ **Do This:** Instruct students to choose one scenario from their activity and discuss with their group what the AI got right, where it needed human support, and what surprised them about how the AI handled the threat.

🔄 **Circulate:** Encourage students to move beyond 'AI is faster' as their key argument. Ask: 'Does AI replace human analysts completely? What tasks might still require human decision-making?' Look for responses that mention AI's adaptability, scalability, and data processing power while recognizing that human oversight is still necessary.

💬 **Discuss:** Click through the animated slide to display the prompts.

- *What patterns did you notice in AI's responses to each threat?*
- *How did AI improve security compared to traditional methods?*

💡 **Discussion Goal:** Students analyze patterns in AI's responses to security threats and evaluate its strengths and weaknesses. Encourage them to consider what AI does well (speed, data analysis, automation) and where it struggles (bias, detecting new types of threats, false positives). Guide them to recognize that AI is a tool, not a replacement for human decision-making in cybersecurity.

- ☑ **Do This:** Explain to students that the next slides will have a statement about AI and cybersecurity. Students will hold a thumbs up for True, thumbs down for False, or thumbs sideways for unsure. Once students have held up their answer, you will reveal the correct answer and explain why.
- ☑ **Do This:** Read the slide and ask students to choose whether it's true or false. Once students have indicated their answer, move onto the next slide to reveal and discuss.
- ☑ **Do This:** Explain why the statement is false: AI is powerful, but it's not perfect—some scams still slip through. Hackers continuously adapt their methods, and AI can sometimes generate false positives or miss subtle attacks. Human judgement should be the first line of defense. Remind students that if they are unsure whether something is a scam, take a step back and cross-check with a trusted source, like speaking to the person who sent the message in person.
- ☑ **Do This:** Read the slide and ask students to choose whether it's true or false. Once students have indicated their answer, move onto the next slide to reveal and discuss.
- ☑ **Do This:** Explain why the statement is false: Social engineering works because it looks convincing—attackers impersonate real people, use familiar logos, and create a sense of urgency. Remind students to always examine the sender details, such as looking to see if the email address matches the school or company's domain. Other ways you can tell are any red flags, like urgent requests, odd grammar, or messages that ask for personal information.
- ☑ **Do This:** Read the slide and ask students to choose whether it's true or false. Once students have indicated their answer, move onto the next slide to reveal and discuss.
- ☑ **Do This:** Explain why the statement is false: Even responding to a scam email can be dangerous—attackers confirm active accounts based on responses and might target students again. Remind students that when in doubt, use this approach: 'STOP-THINK-VERIFY' before interacting with any suspicious message. Never respond to a message with any personal information before verifying that it is actually the person they say they are. The best way to do this is to ask the person in real life.

Wrap Up (5 minutes)

- ☑ **Do This:** Review the learning objectives.
- ☑ **Do This:** Have students take out their The Fabric of the Internet and AI Unit Guide.
- ☑ **Do This:** Place students in pairs or small groups of three.
- ☑ **Do This:** Direct students to briefly share their thoughts about what makes AI effective at enhancing network security and how it compares to traditional methods before individually writing their own response to the Question of the Day in their Unit Guide.

💡 Teaching Tip

Encourage students to go beyond basic thoughts such as “it's faster” by making connections to today's activities.

✔ Assessment Opportunity

The Question of the Day effectively assesses the lesson objectives if students provide responses that explain:

- the role of AI in real-time threat detection.

- An example of AI-based solutions to network security challenges.

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Student Survey



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Lesson 11: Network Security Risk Assessment Project

135 minutes

Overview

Career Journeys

This project-based lesson guides students through conducting a Network Security Risk Assessment, identifying vulnerabilities, and proposing AI-driven security solutions. Students analyze real-world cybersecurity risks, categorize them, and assess their societal and ethical implications before refining their recommendations through peer review. By the end of the project, students communicate their findings in a professional risk assessment report.

Standards

Full Course Alignment

CSTA K-12 Computer Science Standards (2017)

- ▶ **IC** - Impacts of Computing
- ▶ **NI** - Networks & the Internet

CSTA K-12 Computer Science Standards (2026)

- ▶ **SYS** - Systems and Security

Agenda

Day 1 (45 minutes)

Warm Up

Activity

Wrap Up

Day 2 (45 minutes)

Warm Up

Activity

Wrap Up

Day 3 (45 minutes)

Warm Up

Activity

End of Unit Survey

Objectives

Students will be able to:

- Develop and present a Network Security Risk Assessment report
- Evaluate key network vulnerabilities and assess their societal and ethical implications
- Propose AI-driven solutions to mitigate security risks

Preparation

- Print copies of the Network Security Risk Assessment Project Planning Guide (one for each student)

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the students

- **Network Security Risk Assessment Project Planning Guide** - Activity Guide

▼ Make a Copy

Teaching Guide

Day 1 (45 minutes)

How do we create a network security risk assessment?

Warm Up (5 minutes)

 **Discuss:** Click through the animated slide to display the prompts. *What vulnerabilities would you look for first, and why? How might these weaknesses affect the company if exploited?*

- *What vulnerabilities would you look for first, and why?*
- *How might these weaknesses affect the company if exploited?*

 **Discussion Goal:** Students should explore both technical and human vulnerabilities, considering how cybercriminals exploit weak security measures. Guide students to discuss the real-world impact of these breaches on businesses and individuals, including financial loss, reputational damage, and privacy concerns.

 **Do This:** Introduce the Question of the Day

Activity (30 minutes)

 **Distribute:** Give each student a copy of the Network Security Risk Assessment Project Planning Guide. This will guide them through all three days of the project.

 **Do This:** Explain to students that for the next three days, they will work as network security analysts. This means they will select a company scenario, identify vulnerabilities, assess their societal and ethical impacts, and propose both traditional and AI-driven solutions. Along the way, they will refine their recommendations through peer review.

 **Do This:** Click through the animated slide to explain the project requirements and answer any questions.

 **Do This:** Direct students to read the project rubric and hold on to any questions they may have.

 **Discuss:** Click through the animated slide to display the prompts.

- *What do you notice about the project rubric?*
- *What does success look like for this project?*

 **Discussion Goal:** Students share what they notice about the project requirements and rubric and identify the tasks they will need to complete for this project.

 **Do This:** Walk through the first example to model what quality assessment looks like for Critical Vulnerabilities and how it connects to the rubric.

 **Do This:** Walk through the example of societal / ethical impact risk assessment to model what quality work looks like and how it connects to the rubric.

 **Do This:** Walk through the example of critical vulnerabilities risk assessment to model what quality work looks like and how it connects to the rubric.

 **Discuss:** *What barriers do you anticipate to achieving success on this project?*

 **Discussion Goal:** Students identify barriers they anticipate and share any questions they have about the project.

- ☑ **Do This:** Click through the animated slide to introduce the steps students will take on Day 1.
- ☑ **Do This:** Click through the animated slide to introduce the steps students will take on Day 2.
- ☑ **Do This:** Click through the animated slide to introduce the steps students will take on Day 3.
- ☑ **Do This:** Direct students to Lesson 11, Level 1 to choose a company scenario. Instruct them to complete the graphic organizer on their Project Planning Guide for their chosen company.

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Company Case Study Choices

 **Circulate:** As students identify vulnerabilities, check that they are categorizing them correctly (technical, human, systemic). If students list general risks, ask: 'How does this vulnerability actually lead to a security breach?' Encourage deeper thinking by asking: 'What would happen if this vulnerability were exploited? Who would be affected?'

 **Discuss:** Click through the animated slide to display the prompts.

- *What types of vulnerabilities came up the most?*
- *Which ones do you think pose the biggest risks, and why?*

 **Discussion Goal:** Students should identify patterns in network vulnerabilities across different scenarios and evaluate which pose the most significant risks. Guide them to justify their reasoning using examples, considering which vulnerabilities are most easily exploited, which have the greatest impact, and which are hardest to mitigate.

Wrap Up (10 minutes)

 **Discuss:** *How do we create a network security risk assessment?*

- Use the Modified Whip Around protocol to facilitate this discussion.

 **Discussion Goal:** Students explain the key components of a Network Security Risk Assessment, including examples from their own learning from the day's lesson.

Teaching Tip

Modified Whip Around Protocol:

- Individually: Students answer the Question of the Day in 1-2 sentences.
- Small Groups: Students share their responses in groups of 4.
- Whole-Class: 2-3 groups are called on to share key insights or common patterns from their discussion.

Day 2 (45 minutes)

How can we ensure our security recommendations are effective?

Warm Up (5 minutes)

 **Discuss:** *What kind of assistance would you hope AI could provide?*

 **Discussion Goal:** Activate prior knowledge by encouraging students to brainstorm potential AI applications in cybersecurity before being introduced to specific AI-driven solutions.

 **Do This:** Introduce the Question of the Day.

Activity (30 minutes)

 **Do This:** Click through the animated slide to review the steps students will take on Day 2.

 **Do This:** Direct students to Lesson 11, Level 2 to design an AI-powered security tool using AI Chat. Explain that the goal is to create a realistic, responsible AI tool that addresses the vulnerabilities identified on Day 1.



AI-Driven Solutions

 **Circulate:** As students work in AI Chat, listen for surface-level or overly optimistic AI designs (e.g., “This AI stops all hackers”). Prompt deeper thinking with questions like:

- “What would this AI need to know to detect that threat?”
- “Who might be affected if this tool makes a mistake?”
- “Does this tool require human oversight?”

If students struggle to connect their tool to their case study, ask: “How does this AI tool directly reduce the vulnerability you identified yesterday?”

Teaching Tip

If students receive vague or generic AI responses, model how to rephrase prompts for clarity (e.g., “Instead of ‘How can AI help?’, try ‘Design an AI tool that protects hospitals from phishing attacks’”).

Remind students that the goal isn’t to accept the AI’s ideas but to analyze, refine, and improve them for accuracy and ethics.

 **Do This:** Direct students to begin drafting their Network Security Risk Assessment using their Project Planning Guide. Remind them to include their AI security tool design in their draft and encourage students to focus on clear, specific details about their chosen scenario, vulnerabilities, mitigation strategies, and societal impacts.

 **Circulate:** As students are working, check for depth and specificity and that they are clearly connecting their AI tool to the vulnerability it’s meant to solve?

Ask probing questions like:

- How does your AI tool actually reduce this risk?
- Who benefits from this design—and who might be left out?
- What would happen if your AI tool failed or made a wrong prediction?

Support students who are unsure how to integrate AI feedback by prompting:

- Which part of your design could the AI help you clarify or improve?

Teaching Tip

Encourage students to complete this section digitally, even if the rest of their project guide is on paper. This

will make it much easier for them to upload their AI tool design to AI Chat later for feedback.

✓ Assessment Opportunity ▲

- Use this draft to gauge students' understanding of network vulnerabilities and AI-driven solution and to provide quick feedback and positive reinforcement.
- As you review student's work, look for:
 - Clear and specific scenario and risks
 - Thoughtful categorization of vulnerabilities
 - Feasible and well-explained mitigation strategies
 - Ethical/societal impact that goes beyond surface-level thinking

Wrap Up (10 minutes)

 **Discuss:** *How can AI enhance network security?*

- Use the Modified Whip Around protocol to facilitate this discussion.

 **Discussion Goal:** Students should evaluate how AI-driven security tools improve threat detection and mitigation. Encourage them to compare AI's capabilities to traditional security methods and consider how AI can support (but not replace) human decision-making in cybersecurity. Guide students to connect these insights to real-world applications in their risk assessments.

Day 3 (45 minutes)

What makes a network risk assessment thorough and actionable?

Warm Up (5 minutes)

 **Discuss:** Click through the animated slide to display the prompts.

- *What have you learned?*
- *How has your understanding of network vulnerabilities and solutions changed?*

 **Discussion Goal:** Students reflect on their learning by identifying key takeaways about network vulnerabilities, the consequences of those vulnerabilities, mitigation strategies, and AI-driven solutions.

☒ **Do This:** Introduce the Question of the Day

Activity (30 minutes)

☒ **Do This:** Direct students to Lesson 11, Level 3 to use AI Chat to evaluate their AI security tool designs.

 **Circulate:** As students work, check that they are interpreting AI feedback critically, not just copying it. Ask guiding questions like:

- “Do you agree with the AI's feedback? Why or why not?”
- “How can you apply this feedback to make your design more realistic or ethical?”

- “Did the AI notice something you hadn’t considered?” If students receive overly general or incorrect feedback, prompt them to rephrase with specificity (e.g., “Explain exactly how my AI tool addresses the vulnerability I described”).



AI Chat Evaluation

💡 Teaching Tip

Remind students that the focus is on managing AI—using its feedback responsibly, questioning its accuracy, and deciding what to revise based on their own judgment.

☒ **Do This:** Direct students to navigate to Lesson 11, Level 4 in order to complete their final Network Security Risk Assessment. Once students have finished, make sure that they prepare a 30-second oral summary to share with a classmate.

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☒ **Do This:** Divide students into two equal groups and arrange them into two lines, facing each other. Explain the students that they will take turns delivering a 30-second summary of their project and receiving feedback from the peer they are facing. Direct students to use the prompts on the screen to guide their responses if needed, and listen carefully to your partner’s insights. After 5 minutes, they will switch roles. After both pairs have shared, one line will rotate down and they will share again with a new partner.

💡 Teaching Tip

Consider modeling a strong 30-second summary before students begin, demonstrating concise and clear communication, and reiterating the need to answer the displayed prompts. If students struggle to stay within time limits, use a visual or auditory timer to keep rounds on track. Encourage students to ask follow-up questions rather than just saying “good job” (e.g., “How does your solution compare to real-world cases?”).

✔ Assessment Opportunity

These presentations can serve as an informal formative assessment opportunity. As students talk, check for the following:

- Are students addressing all key prompts?
- Do they provide specific details rather than vague answers?
- Are they engaging with feedback and refining their explanations?

End of Unit Survey

After completing this project, students should complete the End of Unit Survey linked in the final level. Their responses help us understand their experiences and improve future versions of this unit.



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